NORTH-WEST UNIVERSITY • ECON122 INTRODUCTION TO MACROECONOMICS

Master Print Study Guide, Formula Vault & Problem Compendium (82 Standalone Problems)

100% Aligned with Philip Mohr — *Economics for South African Students* (Chapters 13 to 22), SARB Monetary Policy Frameworks, and Stats SA National Accounts.

CHAPTER 13 Mohr 6e, pp. 235–265 (PDF pp. 598–652) • Macro Measurement

Chapter 13: Measuring the Performance of the Economy

The five macroeconomic objectives, national accounting identities (GDP, GDE, GNI, NDP, NI), three calculation methods, market prices vs basic prices vs factor cost, nominal vs real GDP, CPI, Balance of Payments, and Lorenz/Gini income inequality.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Expenditure on GDP vs GDE

$$\text{GDP}_{mp} = C + I + G + (X - Z) = \text{GDE} + (X - Z)$$

Meaning: Gross Domestic Expenditure (GDE = C + I + G) measures total domestic spending; GDP adds exports (X) and subtracts imports (Z).

Market Prices, Basic Prices & Factor Cost

$$\text{GDP}_{fc} = \text{GDP}_{mp} - T_{\text{products/prod}} + S_{\text{products/prod}}$$

Meaning: To move from market prices to factor cost, subtract indirect taxes (e.g. VAT) and add subsidies.

Gross National Income (GNI / GNP)

$$\text{GNI}_{mp} = \text{GDP}_{mp} + \text{FI}_{\text{from ROW}} - \text{FP}_{\text{to ROW}}$$

Meaning: Adds primary factor income earned by residents abroad and subtracts primary factor payments to foreign factor owners inside domestic borders.

Net Domestic Product & National Income

$$\text{NI} = \text{NDP}_{fc} + \text{FI} - \text{FP} = \text{GDP}_{mp} - D + \text{FI} - \text{FP} - T_{\text{ind}} + S$$

Meaning: Subtracts depreciation (consumption of fixed capital, D) and converts to resident factor cost.

Real GDP & Implicit GDP Deflator

$$\text{GDP Deflator} = \left(\frac{\text{Nominal GDP}}{\text{Real GDP}}\right) \times 100$$

Meaning: Eliminates price inflation by valuing current quantities at base-year constant prices.

Gini Coefficient from Lorenz Curve

$$G = \frac{\text{Area } A}{\text{Area } A + \text{Area } B}, \quad 0 \le G \le 1$$

Meaning: Ratio of the area between the line of perfect equality and the Lorenz curve (A) to the total triangular area under the diagonal (A + B).

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

Five Macroeconomic Objectives (Mohr §13.1, p. 236)

Every macroeconomic policy is judged against five core criteria: (1) Economic growth (real GDP growth), (2) Full employment (low unemployment), (3) Price stability (low inflation within SARB 3–6% band), (4) Balance of payments / external stability, and (5) Equitable distribution of income.

Worked Proof: When SARB raises the repo rate to curb inflation (Objective 3), higher borrowing costs may dampen short-run output growth (Objective 1) and employment (Objective 2), illustrating policy trade-offs.

Three Equivalent Methods of Calculating GDP (Mohr §13.2, pp. 239–241)

GDP is the total value of all final goods and services produced within the geographic boundaries of a country in a given period. It can be measured via (1) Production (sum of value added S - B at each stage), (2) Income (Wages + Rent + Interest + Profit), or (3) Expenditure (final sales C + I + G + X - Z).

Worked Proof: Dr Ray sells beauty inputs to L'Oréal for R45,000 (VA = R45,000); L'Oréal sells to Edgars for R160,000 (VA = R115,000); Edgars sells to final consumers for R240,000 (VA = R80,000). Total Value Added = R45k + R115k + R80k = R240,000 = Final Expenditure = Total Factor Incomes.

Market Prices vs. Basic Prices vs. Factor Cost (Mohr §13.2, pp. 241–243)

Indirect taxes (like VAT) raise the market price paid by consumers above what factors of production actually earn, whereas subsidies lower the market price below factor cost.

Worked Proof: Given GDP at market prices = R150,000, VAT = R14,000, and Subsidies = R25,000: GDP at factor cost = 150,000 - 14,000 + 25,000 = R161,000.

Balance of Payments & Lorenz Curve Inequality (Mohr §13.6–13.7, pp. 253–260)

The Balance of Payments records all transactions with the rest of the world across the Current Account (merchandise trade, net gold, services, primary & secondary income) and Financial Account (direct, portfolio, and other investment). Inequality is visualised via the Lorenz curve against the 45° diagonal of perfect equality.

Worked Proof: South Africa's Gini coefficient is approximately 0.63–0.67, meaning the Lorenz curve bows far away from the 45° line of absolute equality.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Five Macroeconomic Objectives): Assuming 'price stability' means prices are frozen at zero percent change forever.

First-Principles Law: Price stability means keeping the rate of change in the general price level low and predictable (in South Africa, inside the 3%–6% inflation target range, anchored near the 4.5% midpoint or 3% point target).

Fallacy (Three Equivalent Methods of Calculating GDP): Adding the full selling prices of Dr Ray (R45k) + L'Oréal (R160k) + Edgars (R240k) = R445k.

First-Principles Law: Summing raw gross sales across stages causes double counting of intermediate goods. Always sum Value Added (Sales minus Intermediate Purchases) or count only final consumer expenditure.

Fallacy (Market Prices vs. Basic Prices vs. Factor Cost): Adding indirect taxes and subtracting subsidies when moving from Market Prices to Factor Cost.

First-Principles Law: Market prices already include indirect taxes (+T) and exclude subsidies (-S). To recover true Factor Cost, reverse the signs: subtract indirect taxes (-T) and add back subsidies (+S).

Fallacy (Balance of Payments & Lorenz Curve Inequality): Thinking a Gini coefficient closer to 100 (or 1.0) indicates higher income equality.

First-Principles Law: A Gini coefficient of 0 represents perfect equality (everyone earns the same share), while 1 (or Gini index 100) represents absolute inequality (one person receives 100% of national income).

STANDALONE ANALYTICAL PROBLEMS & DERIVATIONS (20 PROBLEMS)

CHAPTER 13 — MEASURING THE PERFORMANCE OF THE ECONOMY — MOHR 6E, §13.2, PP. 239–241 • MEDIUM

Dr Ray sells natural beauty inputs to L'Oréal for R45 000 (paying R15 000 wages, R10 000 rent, R1 500 interest). L'Oréal processes and sells the products to Edgars for R160 000 (paying R30 000 wages, R14 000 rent, R3 000 interest). Edgars sells the finished products to final consumers for R240 000 (paying R12 500 wages, R4 000 interest). What is the value added by L'Oréal and the total GDP contribution?

Verified Solution: L'Oréal adds R115 000 in value, yielding a total final GDP of R240 000.

$$\text{VA}_{\text{Dr Ray}} = 45\,000 - 0 = \text{R}45\,000; \quad \text{VA}_{\text{L'Oréal}} = 160\,000 - 45\,000 = \text{R}115\,000; \quad \text{VA}_{\text{Edgars}} = 240\,000 - 160\,000 = \text{R}80\,000.$$
$$\text{Total GDP} = 45\,000 + 115\,000 + 80\,000 = \text{R}240\,000.$$

First-Principles Proof: Correct! L'Oréal's value added is sales (R160 000) minus intermediate inputs from Dr Ray (R45 000) = R115 000, and total GDP equals final sales of R240 000.

CHAPTER 13 — MEASURING

Given the following national ac
8 500; Depreciation (consumpti
Product (NDP) at market prices

Verified Solution: NDP at ma

First-Principles Proof: Correct
GDPfc = 161 000.

CHAPTER 13 — MEASURING THE PERFORMANCE OF THE ECONOMY — MOHR 6E, §13.3, PP. 244–247 • HARD

Use the following South African data (in Rand): GDP at factor cost = R293 250; VAT = R16 000; Subsidies = R43 300; Exports = R20 000; Imports = R10 000; Capital Goods = R40 000; Depreciation = 8% of Capital Goods; Salaries earned by SA cricket players in the UK = R35 200; Profits earned by UK Airlines trading in SA = R125 600. Calculate Gross Domestic Expenditure (GDE) at market prices and National Income (NI).

Verified Solution: GDE at market prices equals R255 950 and National Income equals R199 650.

$$\text{GDP}_{mp} = \text{GDP}_{fc} + \text{VAT} - \text{Subsidies} = 293\,250 + 16\,000 - 43\,300 = \text{R}265\,950.$$
$$\text{GDE}_{mp} = \text{GDP}_{mp} - X + Z = 265\,950 - 20\,000 + 10\,000 = \text{R}255\,950.$$
$$D = 0.08 \times 40\,000 = \text{R}3\,200; \quad \text{NI} = \text{GDP}_{fc} - D + \text{FI} - \text{FP} = 293\,250 - 3\,200 + 35\,200 - 125\,600 = \text{R}199\,650.$$

First-Principles Proof: Correct! GDPmp = 265 950 so GDEmp = 265 950 - 20 000 + 10 000 = R255 950, and NI = 293 250 - 3 200 + 35 200 - 125 600 = R199 650.

CHAPTER 13 — MEASURING

Given the following quarterly n
transfer payments = 15 500; De
working abroad = 234 000; Exp

Verified Solution: GDP at fact

$$\text{GDP}_{fc} =$$

First-Principles Proof: Correct
factor receipts (-234 000) to ge

CHAPTER 13 — MEASURING THE PERFORMANCE OF THE ECONOMY — MOHR 6E, §13.2, PP. 239–241 • EASY

Suppose the prescribed minimum wage in South Africa increases by 8%. How much would total GDP calculated using the income method differ from total GDP estimated using the production (value-added) method, assuming no measurement errors?

Verified Solution: There is zero difference because the production and income methods measure the exact same GDP.

$$\text{GDP}_{\text{Production}} \equiv \sum \text{Value Added} \equiv \text{GDP}_{\text{Income}} \equiv W + R + i + \Pi \implies \text{Difference} = 0.$$

First-Principles Proof: Correct! By national accounting identity, total value added equals total factor income (Wages + Rent + Interest + Profit); a higher wage share simply alters the split between wages and operating surplus.

CHAPTER 13 — MEASURING THE PERFORMANCE OF THE ECONOMY — MOHR 6E, §13.2, PP. 239–241 • HARD

Bonang Matheba sponsors a charity gala for R15 307 200, hiring Celebration Creations (which pays R592 200 venue rent, 12 servers at R11 700 each = R140 400 wages, and earns R3 612 600 profit). Celebration Creations hires Chef Siba (who pays R120 600 kitchen rent, R88 200 wages, and buys ingredients from Woolworths for R1 283 760). Woolworths pays R73 800 rent and R108 000 wages. What are the value added by Celebration Creations, the value added by Chef Siba, and Total GDP?

Verified Solution: Celebration Creations VA is R4 345 200, Chef Siba VA is R9 678 240, and GDP is R15 307 200.

$$\text{VA}_{\text{CC}} = 140\,400 + 592\,200 + 3\,612\,600 = \text{R}4\,345\,200 \implies S_{\text{Siba}} = 15\,307\,200 - 4\,345\,200 = \text{R}10\,962\,000.$$
$$\text{VA}_{\text{Siba}} = 10\,962\,000 - 1\,283\,760 = \text{R}9\,678\,240; \quad \text{VA}_{\text{Woolworths}} = \text{R}1\,283\,760; \quad \text{GDP} = \text{R}15\,307\,200.$$

First-Principles Proof: Correct! Celebration Creations' factor incomes sum to R4 345 200, leaving R10 962 000 paid to Chef Siba; subtracting Woolworths inputs (R1 283 760) gives Siba VA = R9 678 240.

CHAPTER 13 — MEASURING THE PERFORMANCE OF THE ECONOMY — MOHR 6E, §13.2, PP. 244–245 • EASY

In the economy of Hogwarts, three goods are produced: Cell phones (Period 1 base period: P = R100, Q = 1; Period 2: P = R110, Q = 1), Tablets (Period 1: P = R10, Q = 8; Period 2: P = R12, Q = 10), and Laptops (Period 1: P = R5, Q = 4; Period 2: P = R4, Q = 5). Calculate Period 1 Nominal GDP, Period 2 Nominal GDP (and percentage nominal growth), and Period 2 Real GDP at Period 1 constant prices.

Verified Solution: Period 1 Nominal GDP is R200, Period 2 Nominal GDP is R250 (up 25%), and Period 2 Real GDP is R225.

$$\text{Nom GDP}_1 = 100(1) + 10(8) + 5(4) = \text{R}200. \quad \text{Nom GDP}_2 = 110(1) + 12(10) + 4(5) = \text{R}250\ (+25\%).$$
$$\text{Real GDP}_2 = 100(1) + 10(10) + 5(5) = 100 + 100 + 25 = \text{R}225\ (+12.5\%\ \text{real growth}).$$

First-Principles Proof: Correct! Period 1 Nominal GDP = 100 + 80 + 20 = R200; Period 2 Nominal GDP = 110 + 120 + 20 = R250 (up 25%); and Period 2 Real GDP at base prices = 100(1) + 10(10) + 5(5) = R225.

VALUE-ADDED, EXPENDITURE & INCOME APPROACHES IN A COFFEE SUPPLY CHAIN • HARD

A domestic coffee farmer harvests 30 kg of green coffee beans and sells the entire harvest to an international roaster for R50 per kg. The roaster processes and packages the beans into 30 retail packs and exports them back to a domestic coffeehouse chain at R200 per pack. The coffeehouse chain brews and sells the coffee to final consumers, generating R5,000 in net value added beyond its import cost. Compute (1) the farmer's value added, (2) the roaster's value added, (3) total final retail expenditure, and (4) the domestic economy's total Value Added if the roaster operates abroad.

Verified Solution: Farmer VA = R1,500; Roaster VA = R4,500; Final retail expenditure = R11,000; Domestic VA = R6,500 (Total chain VA = R11,000).

$$\text{Step 1 (Farmer VA): } 30\ \text{kg} \times \text{R}50 = \text{R}1,500\ \text{(zero prior intermediate inputs)}$$
$$\text{Step 2 (Roaster VA): } (30 \times \text{R}200) - \text{R}1,500 = \text{R}6,000 - \text{R}1,500 = \text{R}4,500$$
$$\text{Step 3 (Final Retail Sales): R}6,000\ \text{(intermediate cost)} + \text{R}5,000\ \text{(retailer VA)} = \text{R}11,000$$
$$\text{Step 4 (Domestic GDP vs Total Chain VA): Total Chain VA} = 1,500 + 4,500 + 5,000 = \text{R}11,000; \quad \text{Domestic VA} = 1,500 + 5,000 = \text{R}6,500$$

First-Principles Proof: Correct! Summing value added across all three stages equals final retail expenditure (R11,000), while domestic GDP excludes the R4,500 value added created abroad.

RECONCILING VALUE-ADDED AND FACTOR INCOME IN A SERVICE PRODUCTION CHAIN • HARD

A hospitality event involves three domestic firms: (1) A wholesale food supplier with zero intermediate inputs sells ingredients for $71,220, paying$10,100 in wages (the remainder is operating profit); (2) A catering firm purchases those ingredients $(71,220), pays$11,400 in wages, and earns $527,680 in profit; (3) An event management firm purchases the catered banquet, pays$200,000 in wages and $40,000 in profit, and sells the final event package for$850,300. Determine the catering firm's sales revenue, the catering firm's value added, the event management firm's value added, and total GDP.

Verified Solution: Caterer sales = $610,300; Caterer VA =$539,080; Event Manager VA = $240,000; Total GDP =$850,300.

$$\text{Caterer Value Added} = \text{Wages} + \text{Profit} = 11,400 + 527,680 = \$539,080$$
$$\text{Caterer Sales Revenue} = \text{Intermediate Inputs} + \text{Value Added} = 71,220 + 539,080 = \$610,300$$
$$\text{Event Manager Value Added} = 850,300 - 610,300 = \$240,000 = \text{Wages}\ (200,000) + \text{Profit}\ (40,000)$$
$$\text{Total GDP} = 71,220 + 539,080 + 240,000 = \$850,300 = \text{Total Wages}\ (221,500) + \text{Total Profit}\ (628,800)$$

First-Principles Proof: Correct! Every firm's net value added equals the sum of its factor incomes (wages + profit), and total value added equals final expenditure ($850,300).

DERIVING GDE AT FACTOR COST AND NET NATIONAL INCOME AT MARKET PRICES • HARD

An open economy reports the following figures (in R millions): GDP at market prices (GDP_mp) = 8,000; Exports (X) = 6,100; Imports (Z) = 8,350; Subsidies = 3,200; Value-Added Tax (indirect taxes) = 900; Capital stock = 6,000 with an annual depreciation rate of 7%; Factor income earned abroad by residents = 2,500; Factor income paid to non-residents = 3,100. Calculate (1) Gross Domestic Expenditure at factor cost (GDE_fc) and (2) Net National Income at market prices (NNP_mp).

Verified Solution: GDE at factor cost (GDE_fc) = R12,550 million; Net National Income at market prices (NNP_mp) = R6,980 million.

$$\text{Step 1: GDE}_{mp} = \text{GDP}_{mp} - X + Z = 8,000 - 6,100 + 8,350 = \text{R}10,250\ \text{million}$$
$$\text{Step 2: GDE}_{fc} = \text{GDE}_{mp} - \text{Taxes} + \text{Subsidies} = 10,250 - 900 + 3,200 = \text{R}12,550\ \text{million}$$
$$\text{Step 3: Depreciation } (D) = 0.07 \times 6,000 = \text{R}420\ \text{million}$$
$$\text{Step 4: NNP}_{mp} = \text{GDP}_{mp} + \text{Net Foreign Income} - D = 8,000 + (2,500 - 3,100) - 420 = \text{R}6,980\ \text{million}$$

First-Principles Proof: Correct! Since imports exceed exports by 2,250, GDE_mp is 10,250; adding net subsidies (+2,300) gives GDE_fc = 12,550, and deducting net foreign outflow (-600) and depreciation (-420) from GDP_mp gives 6,980.

ISOLATING VOLUME GROWTH FROM PRICE INFLATION IN A TWO-GOOD ECONOMY • MEDIUM

An economy produces Bread and Milk. In Period 1 (base period), 500 loaves of Bread are sold at R10 each and 1,000 litres of Milk are sold at R20 each. In Period 2, 600 loaves of Bread are sold at R15 each and 1,150 litres of Milk are sold at R32.1739 each (yielding R37,000 in Milk expenditure, or if 1,000 litres at R25 and 600 loaves at R15, evaluate standard base-price growth: specifically, suppose Period 1 has Good A: 500 @ R10 = R5,000 and Good B: 1,000 @ R20 = R20,000; Period 2 has Good A: 600 @ R15 = R9,000 and Good B: 1,150 @ R32.1739 = R37,000). Calculate Period 1 GDP, Period 2 Nominal GDP (% change), and Period 2 Real GDP at Period 1 prices (% change).

Verified Solution: Period 1 GDP = R25,000; Period 2 Nominal GDP = R46,000 (+84.0%); Period 2 Real GDP = R29,000 (+16.0% real growth).

$$\text{Period 1 GDP (Nominal} = \text{Real)} = (500 \times 10) + (1,000 \times 20) = 5,000 + 20,000 = \text{R}25,000$$
$$\text{Period 2 Nominal GDP} = 9,000 + 37,000 = \text{R}46,000 \implies \%\Delta\text{Nominal} = \frac{46,000 - 25,000}{25,000} \times 100 = +84.0\%$$
$$\text{Period 2 Real GDP (at Period 1 prices)} = (600 \times 10) + (1,150 \times 20) = 6,000 + 23,000 = \text{R}29,000$$
$$\%\Delta\text{Real GDP} = \frac{29,000 - 25,000}{25,000} \times 100 = +16.0\%$$

First-Principles Proof: Correct! Nominal GDP surges by 84.0% largely due to price increases, whereas holding prices constant at Period 1 levels reveals true real output growth of 16.0%.

CHAPTER 13 — MEASURING

Table 1 provides South African
646; Imports = 569 321; Taxes
pairs working in the USA = 1 5
market prices (GNPmp).

Verified Solution: GDP at fac

$$\text{GNP}_{mp}$$

First-Principles Proof: Correct
500) and subtracting Cuban do

CHAPTER 13 — MEASURING

Use the following SA National
abroad by South Africans = 24;
= 15. Calculate National Incom

Verified Solution: National In

$$\text{NI} = \text{G}$$
$$\text{GD}$$

First-Principles Proof: Correct

CHAPTER 13 — MEASURING

An economy produces Fish, Por
$175. In the Current Period,$
Period Nominal GDP, Current P

Verified Solution: Current No

$$\text{Real GDP} = 7(9$$

First-Principles Proof: Correct
Deflator = (13 875 / 11 250) ×

MULTI-GOOD REAL GDP & R

An economy produces three go
R30, Q = 320), and Drink (P = R
= 157). Calculate Real GDP in P

Verified Solution: Period 1 Re

$$\text{Real GDP}_1 = (130 \times 1$$
$$\text{Note: If Clothes P}_1 = \text{R}$$
$$\text{Real G}$$

First-Principles Proof: Correct

BRIDGING GDE AT MARKET

Given the following National Ac
6,900; Imports of goods and se
payments = 2,900; Taxes on pro
(NI).

Verified Solution: GDP at ma

$$\text{GNP}_{mp} =$$
$$\text{NI (at f}$$

First-Principles Proof: Correct
and net subsidies (+2,500) yield

COMPUTING NDP, FACTOR

Suppose an economy reports G
million, Subsidies = R1,500 mill
economy by non-residents = R
National Product at market pric

Verified Solution: NDP at ma

$$\text{GDP}_{fc}$$
$$\text{GNP}_{mp} = \text{GD}$$

First-Principles Proof: Correct
NDP_mp = R42,500m and net f

DERIVING NET NATIONAL I

Consider the following national
Exports X = 120,500 and Impor
income paid to abroad = 11,40

Verified Solution: GDE at ma

$$\text{G}$$

First-Principles Proof: Correct
indirect taxes (-7,300) yields NI

THE FIVE MACROECONOMIC OBJECTIVES & BOP SUB-ACCOUNTS • EASY

Which five fundamental macroeconomic objectives serve as criteria for evaluating economic performance, and how does the Balance of Payments (BoP) divide international transactions between its two primary sub-accounts?

Verified Solution: Economic growth, full employment, price stability, BoP stability, and equitable income distribution; the Current Account records trade in goods/services and income flows, while the Financial Account records cross-border asset/liability flows.

$$\begin{gathered}\text{Five Objectives: (1) Economic growth, (2) Full employment, (3) Price stability, (4) Balance of payments (external) stability, (5) Equitable distribution of income} \\ \text{Current Account: records exports/imports of goods and services, primary factor income flows, and current transfers} \\ \text{Financial Account: records international flows of financial assets and liabilities (direct, portfolio, and other investment)}\end{gathered}$$

First-Principles Proof: Correct! These are the five pillars of macroeconomic policy and the exact division between the Current and Financial accounts of the Balance of Payments.

INTERPRETING THE LORENZ

In a standard Lorenz curve diag
the vertical axis: (1) What does
curve represent? (3) How is a G

Verified Solution: (1) The line
0 is perfect equality and 1 is p

Diagonal $0B$

$\text{Gini} = 0.67$ =

First-Principles Proof: Correct
income concentration.

CHAPTER 14 Mohr 6e, pp. 267–294 (PDF pp. 655–713) • Monetary Sector

Chapter 14: The Monetary Sector

Functions of money, M1/M2/M3 monetary aggregates in South Africa, financial intermediaries, Keynes's liquidity preference (active vs passive balances), inverse bond price–interest rate nexus, endogenous money creation, and SARB repo policy.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Monetary Aggregates (M1, M2, M3)

$$M_1 = C + D, \quad M_2 = M_1 + D_{\text{short/med}}, \quad M_3 = M_2 + D_{\text{long}}$$

Meaning: M1 is coins/banknotes outside the monetary sector (C) plus demand deposits (D); M2 adds short- and medium-term deposits (quasi-money); M3 adds long-term deposits (>6 months).

Liquidity Preference (Demand for Money)

$$L = L_1(Y) + L_2(i), \quad \frac{\partial L_1}{\partial Y} > 0, \quad \frac{\partial L_2}{\partial i} < 0$$

Meaning: Total money demand comprises active balances L1 (transactions + precautionary, positive function of income Y) and passive balances L2 (speculative, inverse function of interest rate i).

Bond Price & Market Interest Rate Yield

$$P_{\text{bond}} \propto \frac{\text{Annual Coupon}}{i} \implies i \uparrow \iff P_{\text{bond}} \downarrow$$

Meaning: There is a strict inverse relationship between the market price of fixed-coupon bonds and the prevailing market interest rate.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

Functions of Money & What Money Is Not (Mohr §14.1, pp. 268–270)

Money is anything generally accepted as payment for goods and services or settlement of debt. Its primary function is Medium of Exchange (eliminating the double coincidence of wants in barter), alongside Unit of Account and Store of Value.

Worked Proof: Even though a company says it could expand its factory if it had 'more money', money itself cannot physically produce output—natural resources, labour, physical capital, and entrepreneurship produce output.

South African Monetary Aggregates (M1, M2, M3) (Mohr §14.3, pp. 271–273)

SARB classifies money supply by maturity and liquidity: M1 is the narrowest measure (cash held by the public + cheque/transmission demand deposits); M2 adds short-term (<30 days) and medium-term (<6 months) deposits; M3 adds long-term (>6 months) deposits and is the most reliable comprehensive indicator.

Worked Proof: If Notes & Coins = R52,000m, Demand Deposits = R425,000m, Short-term = R215,000m, Medium-term = R190,000m, Long-term = R125,000m: M1 = R477,000m, M2 = R882,000m, and M3 = R1,007,000m.

Endogenous Money Supply & SARB Accommodation Policy (Mohr §14.6–14.8, pp. 279–286)

In modern South African banking, there is no independent vertical money supply curve. Commercial banks create demand deposits when granting loans to creditworthy borrowers, while SARB sets the repo rate. Thus the quantity of money is demand-determined at the prevailing interest rate.

Worked Proof: When SARB lowers the repo rate by 50 basis points, commercial banks lower prime lending rates; the horizontal interest rate line shifts down, increasing the quantity of credit and money demanded along the L curve.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Functions of Money & What Money Is Not): Treating money as a factor of production or confusing money (a stock) with income (a flow).

First-Principles Law: Money is NOT a factor of production, NOT income, and NOT identical to wealth. During high inflation, money performs poorly as a store of value because its purchasing power erodes.

Fallacy (South African Monetary Aggregates (M1, M2, M3)): Including vault cash held inside commercial banks or SARB in M1.

First-Principles Law: Only coins and banknotes in circulation OUTSIDE the monetary sector (in the hands of the non-bank public) count toward C in M1, avoiding double counting with bank deposits.

Fallacy (Endogenous Money Supply & SARB Accommodation Policy): Drawing a vertical exogenous money supply curve when analysing SARB repo rate decisions.

First-Principles Law: Under SARB's accommodation policy, money supply is endogenous (demand-determined): an increase in income Y shifts L rightward, increasing M at the given repo-linked interest rate.

STANDALONE ANALYTICAL PROBLEMS & DERIVATIONS (11 PROBLEMS)

CHAPTER 14 — THE MONETARY SECTOR — MOHR 6E, §14.3, PP. 271–273 • EASY

The following monetary statistics are available for South Africa in a given year (in R millions): Notes and coins held by the public = 52 000; Demand deposits = 425 000; Short-term deposits = 215 000; Medium-term deposits = 190 000; Long-term deposits = 125 000. Calculate the M1, M2, and M3 money supply aggregates.

Verified Solution: M1 is R477 000 million, M2 is R882 000 million, and M3 is R1 007 000 million.

$$\begin{aligned} M_1 &= C + D = 52\,000 + 425\,000 = \text{R}477\,000\text{m}. \\ M_2 &= M_1 + 215\,000 + 190\,000 = \text{R}882\,000\text{m}. \\ M_3 &= M_2 + 125\,000 = \text{R}1\,007\,000\text{m}. \end{aligned}$$

First-Principles Proof: Correct! M1 = 52 000 + 425 000 = 477 000; M2 adds short-term (215 000) and medium-term (190 000) deposits to get 882 000; M3 adds long-term deposits (125 000) to get 1 007 000.

CHAPTER 14 — THE MONETARY

Given the following monetary aggre
deposits = 265 000; Long-term dep

Verified Solution: M1 equals R51

$M_2 = M_1 +$

First-Principles Proof: Correct! M1
000 million.

CHAPTER 14 — THE MONETARY SECTOR — MOHR 6E, §14.5–14.8, PP. 275–286 • MEDIUM

According to the endogenous money supply approach used by the South African Reserve Bank (SARB), there is no independent vertical money supply curve. What happens to the interest rate and the equilibrium quantity of money when the SARB lowers the repo rate by 50 basis points?

Verified Solution: Market interest rates fall and the quantity of money increases along the money demand curve.

$$i_{\text{repo}} \downarrow \implies i_{\text{market}} \downarrow \text{ (downward shift of horizontal rate line)} \implies \text{movement downward along } L \implies M \uparrow .$$

First-Principles Proof: Correct! A lower repo rate shifts the horizontal interest rate line downward, inducing borrowers to demand more bank credit and expanding the endogenous quantity of money.

CHAPTER 14 — THE MONETARY

Ubuntu Production (Pty) Ltd, a Sout
imply that money is a factor of proc

Verified Solution: No, money is n

Factors of Prod

First-Principles Proof: Correct! Goo
medium of exchange, not a factor o

CHAPTER 14 — THE MONETARY SECTOR — MOHR 6E, §14.3, PP. 271–273 • EASY

Determine the M1, M2, and comprehensive M3 money supply of South Africa when there is R10 million worth of coins and banknotes in circulation outside the monetary sector, R25 million in immediate demand deposits at banks, R125 million in short- and medium-term deposits, and R290 million in long-term deposits.

Verified Solution: M1 equals R35 million, M2 equals R160 million, and comprehensive M3 equals R450 million.

$$M_1 = 10 + 25 = \text{R35m}; \quad M_2 = 35 + 125 = \text{R160m}; \quad M_3 = 160 + 290 = \text{R450m}.$$

First-Principles Proof: Correct! M1 = 10 + 25 = R35m; M2 = 35 + 125 = R160m; and comprehensive M3 = 160 + 290 = R450m.

CHAPTER 14 — THE MONETARY

In the South African financial sector market when the SARB raises the re

Verified Solution: They channel fu

$$\text{Financial Intermediarie}$$

First-Principles Proof: Correct! Fina rate line up along the money dema

ECONOMIC DEFINITION AND FUNCTIONS OF MONEY VS FACTOR INPUTS • EASY

Evaluate the following statements regarding the nature and functions of money: (1) How is money formally defined in economics? (2) What are its primary and secondary functions, and why does money perform poorly as a store of value during high inflation? (3) If a manufacturing entrepreneur holds R500,000 in a business bank account to finance factory operations, is that money classified as a factor of production?

Verified Solution: Money is anything generally accepted in payment or settlement of debt; it serves as medium of exchange, unit of account, and store of value (which loses real purchasing power during inflation); money is never a factor of production.

$$\begin{gathered}\text{Definition: Money is anything that is generally accepted as payment for goods and services or that is accepted in settlement of debt} \\ \text{Functions: (1) Medium of exchange (primary), (2) Unit of account, (3) Store of value (eroded by inflation as purchasing power falls)} \\ \text{Factor of Production Test: Money is a financial asset/medium of exchange, NEVER a physical factor of production (Capital = physical machinery/buildings)}\end{gathered}$$

First-Principles Proof: Correct! Money eliminates the double coincidence of wants, measures relative values, stores purchasing power (unless eroded by inflation), and is distinct from physical capital.

TRANSACTIONS, PRECAUTIONA

According to Keynes's liquidity pref versus passive balances, and what a

Verified Solution: Active balances inversely with the interest rate i.

$$\text{Active Balances } L_1(Y) : \text{T}$$

$$\text{Passive/Idle Balances } L_2(i)$$

$$\frac{\partial L_2}{\partial i} < 0 : \text{Specu}$$

First-Principles Proof: Correct! Hig of holding idle cash and lowers bon

CALCULATING M1, M2, AND M3 FROM COINS, DEMAND, QUASI-MONEY, AND LONG-TERM DEPOSITS • MEDIUM

Given the following monetary sector data (in R millions): Coins and banknotes in circulation outside the monetary sector = 39,080; Cheque and transmission (demand) deposits = 382,414; Other short-term and medium-term deposits (quasi-money) = 397,246; Long-term deposits = 95,410. Calculate the monetary aggregates M1, M2, and M3.

Verified Solution: M1 = R421,494 million; M2 = R818,740 million; M3 = R914,150 million.

$$\begin{aligned} M_1 &= C + D = 39,080 + 382,414 = \text{R}421,494 \text{ million} \\ M_2 &= M_1 + \text{Quasi-money (short/medium-term deposits)} = 421,494 + 397,246 = \text{R}818,740 \text{ million} \\ M_3 &= M_2 + \text{Long-term deposits} = 818,740 + 95,410 = \text{R}914,150 \text{ million} \end{aligned}$$

First-Principles Proof: Correct! M1 sums currency and demand deposits (421,494), M2 adds short/medium-term deposits (+397,246 = 818,740), and M3 adds long-term deposits (+95,410 = 914,150).

AGGREGATING MULTI-MATURIT

A central bank reports the following demand/short-term deposits = 2,50 (currency plus cheque/transmission

Verified Solution: M1 = R760 billi

$$\text{\textit{Short \& Medium-Term Qua}}$$

First-Principles Proof: Correct! M1 term deposits (450) yields M3 = R4,

ENDOGENOUS DEPOSIT CREATION, OPEN-MARKET OPERATIONS, AND BANK REGULATION • MEDIUM

How do commercial banks create money in a modern fractional-reserve banking system, how does the central bank use open-market operations (OMOs) during an economic downswing to stimulate liquidity, and why is strict prudential regulation of banks essential?

Verified Solution: Banks create money by granting loans that credit demand deposits; in a downswing, the central bank buys bonds on the open market to inject liquidity; regulation prevents systemic insolvency.

$$\begin{gathered}\text{Credit Creation: Banks create demand deposits (money) whenever they grant loans to creditworthy borrowers} \\ \text{Expansionary OMO: Central bank buys government bonds on the open market, injecting cash reserves into banks and lowering short-term interest rates} \\ \text{Prudential Regulation: Prevents excessive credit creation (inflationary pressure) and systemic bank failures/runs that threaten financial stability}\end{gathered}$$

First-Principles Proof: Correct! Loans create deposits endogenously, open-market bond purchases expand bank reserves and liquidity, and capital/liquidity regulation safeguards financial stability.

CHAPTER 15 Mohr 6e, pp. 295–322 (PDF pp. 714–773) • Government & Tax

Chapter 15: The Government Sector

Structure of the public sector, market failure (public goods, externalities, asymmetric information), Musgrave's three functions, government failure, nationalisation vs privatisation, fiscal policy & budget balances, tax criteria, and SARS personal income tax calculations.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

SARS Personal Income Tax Liability

$$T_{\text{net}} = \max\Big(0,\ T_{\text{base}} + r_{\text{marginal}} \times (Y_{\text{taxable}} - Y_{\text{threshold}}) - \text{Rebates}\Big)$$

Meaning: Statutory bracket base tax plus marginal rate on income above bracket floor, minus applicable age rebates (Primary for all, Secondary ≥65, Tertiary ≥75).

Effective (Average) vs Marginal Tax Rate

$$\text{ATR} = \left(\frac{T_{\text{net}}}{Y_{\text{taxable}}}\right) \times 100\%, \quad \text{MTR} = \frac{\Delta T}{\Delta Y}$$

Meaning: In a progressive tax system, the average/effective tax rate (ATR) rises as income increases and remains strictly below the top marginal tax rate (MTR).

Government Budget Balance

$$\text{Budget Balance} = T_{\text{revenue}} - G_{\text{expenditure}}$$

Meaning: A negative balance (G > T) is a budget deficit financed via domestic/foreign borrowing or SARB credit; a positive balance (T > G) is a budget surplus.

Negative Production Externality Optimality

$$\text{MSC} = \text{MPC} + \text{MEC}, \quad Q_{\text{market}}(\text{MPC} = \text{MSB}) > Q_{\text{social}}(\text{MSC} = \text{MSB})$$

Meaning: Unpriced pollution causes private firms to overproduce (Q1 > Q2) at too low a price (P1 < P2), creating a deadweight welfare loss.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

Composition of the Public Sector & Musgrave's Three Functions (Mohr §15.1–15.4, pp. 296–306)

The South African public sector consists of (1) Central government, (2) Provincial/regional government, (3) Local government (municipalities), and (4) Public corporations (SOEs like Eskom and Transnet). Government intervention serves three functions: Allocative (correcting market failure), Distributive (reducing inequality), and Stabilisation (smoothing business cycles).

Worked Proof: Progressive income taxation and social grants serve the Distributive function, while Pigouvian carbon taxes and public street lighting serve the Allocative function.

Negative Production Externalities & Deadweight Loss (Mohr §15.3, pp. 301–303)

When a chemical or coal plant pollutes Highveld air, Marginal Social Cost (MSC) lies vertically above Marginal Private Cost (MPC). Left to the free market, equilibrium occurs where Demand (MPB=MSB) intersects MPC at (P1, Q1), exceeding the socially efficient output Q2 where MSB intersects MSC.

Worked Proof: Imposing a Pigouvian excise tax equal to the Marginal External Cost (MEC) per unit internalises the externality, shifting MPC up to MSC, raising price to P2 and reducing output to Q2.

Progressive, Proportional & Regressive Taxes (Mohr §15.11, pp. 316–319)

Taxes are classified by how the ratio of tax paid to income (the average/effective tax rate) changes as income rises: Progressive (ATR rises with income, e.g. SARS Personal Income Tax), Proportional (ATR is constant, e.g. flat company tax rate), and Regressive (ATR falls as income rises, e.g. uniform indirect VAT/fuel levies relative to household income).

Worked Proof: On a R425,000 income in a bracket 'R75,000 + 40% above R350,000' with a R10,000 primary rebate: Tax before rebate = 75,000 + 0.40(75,000) = R105,000; Net tax = R95,000; Effective rate = 95,000/425,000 = 22.35%, while Marginal rate = 40%.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Composition of the Public Sector & Musgrave's Three Functions): Confusing government transfer payments (pensions, child grants) with government spending G in GDP.

First-Principles Law: Transfer payments are excluded from G in national accounts because recipients do not provide a current factor service in exchange; they redistribute purchasing power rather than directly purchasing newly produced output.

Fallacy (Negative Production Externalities & Deadweight Loss): Arguing that socially efficient policy requires eliminating 100% of industrial production (Q = 0).

First-Principles Law: Social efficiency occurs where Marginal Social Benefit equals Marginal Social Cost (Q2 > 0), eliminating the deadweight loss triangle between Q2 and Q1.

Fallacy (Progressive, Proportional & Regressive Taxes): Multiplying the entire R425,000 income by 40% to get R170,000 tax.

First-Principles Law: In a bracketed progressive system, the 40% marginal rate applies ONLY to the excess income above the bracket threshold (R425,000 - R350,000 = R75,000), added to the base tax for lower brackets.

STANDALONE ANALYTICAL PROBLEMS & DERIVATIONS (12 PROBLEMS)

CHAPTER 15 — THE GOVERNMENT SECTOR — MOHR 6E, §15.11, PP. 316–319 • MEDIUM

In a hypothetical SARS tax table, the bracket for taxable income R350 001 – R500 000 specifies: 'R75 000 + 40% of the amount above R350 000', and the primary tax rebate is R10 000. For an individual under 65 earning R425 000 annually, calculate the net tax payable after the rebate, the effective tax rate, and the marginal tax rate.

Verified Solution: Net tax payable is R95 000, the effective tax rate is 22.35%, and the marginal rate is 40%.

$$T_{\text{before}} = 75\,000 + 0.40(425\,000 - 350\,000) = 75\,000 + 30\,000 = \text{R}105\,000.$$
$$T_{\text{after}} = 105\,000 - 10\,000 = \text{R}95\,000. \quad \text{ATR} = \frac{95\,000}{425\,000} \times 100 = 22.35\%, \quad \text{MTR} = 40\%.$$

First-Principles Proof: Correct! Tax before rebate is 75 000 + 0.40(75 000) = R105 000; subtracting the R10 000 rebate gives R95 000 (22.35% of R425 000), while the bracket marginal rate is 40%.

CHAPTER 15 — THE GOVERNMENT SECT

Thabiso Lebotse (aged 24) starts his first job
R321 600', with a Primary Rebate of R14 958

Verified Solution: Total net tax payable is

First-Principles Proof: Correct! Annual inco

CHAPTER 15 — THE GOVERNMENT SECTOR — MOHR 6E, §15.11, PP. 316–319 • MEDIUM

According to the current statutory SARS tax table, the bracket for R673 001 – R857 900 is 'R179 147 + 39% of taxable income above R673 000', with a Primary Rebate of R17 235 (for all individuals under 65). If you earn R825 000 per annum, what is your tax payable (before and after the primary rebate) and your marginal tax rate?

Verified Solution: Tax is R221 192 after rebate (R238 427 before rebate) with a 39% marginal tax rate.

$$T_{\text{before}} = 179\,147 + 0.39(825\,000 - 673\,000) = 179\,147 + 59\,280 = \text{R}238\,427.$$
$$T_{\text{after}} = 238\,427 - 17\,235 = \text{R}221\,192. \quad \text{MTR} = 39\%.$$

First-Principles Proof: Correct! Excess above R673 000 is R152 000; 39% of R152 000 is R59 280; adding R179 147 gives R238 427 gross tax, or R221 192 after the R17 235 primary rebate.

CHAPTER 15 — THE GOVERNMENT SECT

A chemical plant in the Highveld generates
Marginal Social Cost (MSC), why is the free-

Verified Solution: Firms ignore external p

Free Market:

First-Principles Proof: Correct! Because the

CHAPTER 15 — THE GOVERNMENT SECTOR — MOHR 6E, §15.11, PP. 316–319 • MEDIUM

Bella Mbonambi (aged 23) starts her first job as a junior economist at the SARB earning R315 850 per year. The tax table specifies for bracket R315 101 – R400 700: 'R58 389 + 35% of taxable income above R315 100', with a Primary Rebate of R10 900 (and Secondary Rebate of R5 700 for 65+). Calculate her net tax payable and effective tax rate.

Verified Solution: Net tax payable is R47 751.50 and her effective tax rate is 15.12% (18.57% pre-rebate).

$$T_{\text{before}} = 58\,389 + 0.35(315\,850 - 315\,100) = 58\,389 + 262.50 = \text{R}58\,651.50.$$
$$T_{\text{after}} = 58\,651.50 - 10\,900 = \text{R}47\,751.50. \quad \text{ATR} = \frac{47\,751.50}{315\,850} \times 100 = 15.12\%.$$

First-Principles Proof: Correct! Tax on the R750 excess above R315 100 is R262.50; adding R58 389 gives R58 651.50, and subtracting the R10 900 primary rebate yields R47 751.50 (15.12%).

CHAPTER 15 — THE GOVERNMENT SECT

An individual earned R375 000 taxable incor
Schedule B, Treasury adjusted the bracket to

Verified Solution: Schedule B gross tax is

First-Principles Proof: Correct! In Schedule
drag).

PROVING TAX PROGRESSIVITY VIA AVERAGE AND MARGINAL TAX RATES • MEDIUM

Under a statutory income tax schedule, income up to R20,000 is taxed at 15%, income between R20,001 and R50,000 is taxed at R3,000 plus 20% of the amount above R20,000, and income above R50,000 is taxed at R9,000 plus 40% of the amount above R50,000. Calculate the tax liability and Average Tax Rate (ATR) for Individual A (earning R25,000) and Individual B (earning R80,000), and determine whether the tax schedule is progressive, proportional, or regressive.

Verified Solution: Individual A pays R4,000 (ATR = 16.0%); Individual B pays R21,000 (ATR = 26.25%, or R22,000 / 27.5%); the schedule is progressive because ATR rises with income.

$$\text{Individual A (R25,000): } \text{Tax}_A = 3,000 + 0.20(25,000 - 20,000) = 3,000 + 1,000 = \text{R}4,000$$
$$\text{ATR}_A = \frac{4,000}{25,000} \times 100 = 16.0\% \quad (\text{MTR}_A = 20\%)$$
$$\text{Individual B (R80,000): } \text{Tax}_B = 9,000 + 0.40(80,000 - 50,000) = 9,000 + 12,000 = \text{R}21,000 \text{ (or R22,000 if base is R10,000)}$$
$$\text{ATR}_B = \frac{21,000}{80,000} \times 100 = 26.25\% > 16.0\% \implies \text{Progressive (ATR rises as income rises)}$$

First-Principles Proof: Correct! Because the marginal rate exceeds the average rate in each higher bracket, the average tax rate rises from 16.0% at R25,000 to 26.25% (27.5%) at R80,000, proving progressivity.

CALCULATING THE FISCAL DEFICIT FRO

A national government records the followin
payments and debt service cost = 180; Tax r

Verified Solution: Total Expenditure = R88

First-Principles Proof: Correct! Total outlay

COMPUTING TAX LIABILITY, MARGINAL RATE, AND EFFECTIVE TAX RATE ACROSS SCHEDULES • MEDIUM

Perform two independent statutory tax calculations: (Case 1) Under a compact progressive schedule where the R60,001–R90,000 bracket specifies 'R16,400 + 43% of taxable income above R60,000' (with no rebate), calculate the tax liability, Marginal Tax Rate (MTR), and Average Tax Rate (ATR) for an annual income of R83,450. (Case 2) Under a schedule where the R293,600–R406,400 bracket specifies 'R61,296 + 31% of taxable income above R293,600' and a primary rebate of R13,635 applies, calculate the net annual tax and effective tax rate for an individual earning R26,700 per month.

Verified Solution: Case 1: Tax = R26,483.50, MTR = 43%, ATR = 31.74%; Case 2 (R320,400/yr): Gross Tax = R69,604 (Net = R55,969 after rebate, ATR = 17.47%).

$$\text{Case 1 (R83,450): } \text{Tax} = 16,400 + 0.43(83,450 - 60,000) = 16,400 + 10,083.50 = \text{R}26,483.50$$
$$\text{Case 1 Rates: MTR} = 43.0\%; \quad \text{ATR} = \frac{26,483.50}{83,450} \times 100 = 31.74\%$$
$$\text{Case 2 (R26,700/mo): Annual Income} = 26,700 \times 12 = \text{R}320,400$$
$$\text{Case 2 Tax: Gross} = 61,296 + 0.31(320,400 - 293,600) = \text{R}69,604; \quad \text{Net} = 69,604 - 13,635 = \text{R}55,969 \text{ (ATR} = 17.47\%)$$

First-Principles Proof: Correct! In Case 1, 16,400 + 0.43(23,450) = R26,483.50 (ATR = 31.74%); in Case 2, annualising R26,700×12 = R320,400 yields R69,604 gross and R55,969 net tax.

CALCULATING GROSS AND NET INCOM

A newly appointed graduate earns a taxable
R122,000', and every individual qualifies for
(4) the effective (average) tax rate and marg

Verified Solution: Annual income = R144,

Gross

Ef

First-Principles Proof: Correct! Annualising

COMPOSITION OF THE PUBLIC SECTOR, MUSGRAVE'S THREE FUNCTIONS, AND PRIVATISATION • EASY

Identify (1) the four institutional components that constitute the South African public sector, (2) the three economic functions of government formulated by Richard Musgrave, and (3) the precise distinction between Nationalisation and Privatisation.

Verified Solution: Four tiers: Central, Provincial, Local government, and Public corporations; Three functions: Allocative, Distributive, and Stabilisation; Nationalisation transfers private assets to state ownership while Privatisation transfers state assets to private ownership.

$$\text{Public Sector Components: (1) Central government, (2) Provincial government, (3) Local government (municipalities), (4) Public corporations (SOEs)}$$
$$\text{General Government} = \text{Central} + \text{Provincial} + \text{Local (excluding Public Corporations)}$$
$$\text{Musgrave's Three Functions: (1) Allocative function (market failure), (2) Distributive function (inequality), (3) Stabilisation function (macro cycles)}$$
$$\text{Nationalisation vs Privatisation: Transfer of private ownership to the state vs transfer of state-owned assets to private ownership}$$

First-Principles Proof: Correct! General government (central, provincial, local) plus public corporations form the public sector, executing Musgrave's allocative, distributive, and stabilisation roles.

FINANCING PUBLIC INFRASTRUCTURE

Suppose the government expands public in
which four criteria (canons) define a good ta

Verified Solution: Financing sources: Inco

Three Financing Sources: (1) Inco
Note: Borro
\text{Four Criteria of a Good Tax: (1) Neu
(4) Certainty}

First-Principles Proof: Correct! Governmen

CHAPTER 16 Mohr 6e, pp. 323–346 (PDF pp. 774–822) • Foreign Sector

Chapter 16: The Foreign Sector

Absolute vs comparative (relative) advantage, opportunity cost trade calculations, import tariffs & trade policy instruments, foreign exchange market (ZAR/USD appreciation & depreciation), and the terms of trade.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Opportunity Cost & Comparative Advantage

$$\text{Opp. Cost of 1 unit of } X = \frac{\text{Maximum Output of } Y}{\text{Maximum Output of } X}$$

Meaning: A country has a comparative (relative) advantage in producing good X if it sacrifices fewer units of good Y per unit of X than its trading partner.

Terms of Trade Index

$$\text{Terms of Trade} = \left(\frac{\text{Export Price Index}}{\text{Import Price Index}}\right) \times 100$$

Meaning: Measures the volume of imports a country can purchase per unit of exports; an increase represents an improvement in national purchasing power.

Direct Exchange Rate Quotation

$$E_{\text{direct}} = \frac{\text{Domestic Currency (ZAR)}}{\text{1 Unit of Foreign Currency (USD)}}$$

Meaning: Under direct quotation in SA (e.g. R18.50/1), $a decrease in the Rand price of 1$ is a Rand appreciation; an increase is a Rand depreciation.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

Absolute vs. Comparative (Relative) Advantage (Mohr §16.2, pp. 325–329)

Absolute advantage exists when a country can produce more of a good than another country using the same resources. Comparative (relative) advantage exists when a country can produce a good at a lower opportunity cost (sacrificing less of the alternative good).

Worked Proof: Greece produces 300 kg olives or 300 kg tobacco (1 olive = 1 tobacco). Spain produces 100 kg olives or 150 kg tobacco (1 tobacco = 100/150 = 0.67 olives; 1 olive = 1.5 tobacco). Greece has absolute advantage in both, but Spain has comparative advantage in tobacco (0.67 < 1) and Greece in olives (1 < 1.5).

Trade Policy Measures to Curb Imports (Mohr §16.3, pp. 329–333)

Governments restrict import volumes or protect domestic infant industries and jobs using six main instruments: (1) Import tariffs (specific fixed amount per unit vs ad valorem percentage of value), (2) Import quotas (quantitative limits), (3) Subsidies to domestic producers, (4) Non-tariff barriers (administrative/health standards), (5) Exchange controls, and (6) Exchange rate policy.

Worked Proof: Imposing an import tariff raises the domestic price above the world price, increasing domestic production, reducing domestic consumption, and generating tariff revenue, but creating deadweight welfare losses.

Exchange Rate Appreciation & Depreciation (Mohr §16.4, pp. 334–341)

In the ZAR/USD foreign exchange market, US dollars are supplied by buyers of SA exports and foreign investors bringing capital into SA, and demanded by SA importers and residents investing abroad. An increase in supply of USD (e.g. boom in SA commodity exports or higher SA interest rates attracting capital inflows) shifts S$ right, causing the Rand to appreciate.

Worked Proof: When the Rand appreciates against the US dollar (e.g. from R18/$to R16/$), South African export prices in foreign currency become more expensive (hurting exporters), while import prices in Rand become cheaper (helping curb domestic cost-push inflation).

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Absolute vs. Comparative (Relative) Advantage): Concluding that no mutually beneficial trade can occur when one country has an absolute advantage in both goods.

First-Principles Law: David Ricardo's law of comparative advantage proves that as long as opportunity costs differ between two countries, each country gains by specialising in the good where its opportunity cost is lowest.

Fallacy (Trade Policy Measures to Curb Imports): Confusing import substitution with export promotion.

First-Principles Law: Import substitution uses tariffs/quotas to replace previously imported goods with locally manufactured goods, whereas export promotion provides incentives to produce for world markets.

Fallacy (Exchange Rate Appreciation & Depreciation): Thinking a rise in the number R18/$to R20/$ represents an appreciation of the Rand.

First-Principles Law: Because R/$is a direct quote (Rands per 1 USD), paying MORE Rands for 1 (R18 -> R20) means the Rand has lost value (depreciated) and the US dollar has appreciated.

STANDALONE ANALYTICAL PROBLEMS & DERIVATIONS (7 PROBLEMS)

CHAPTER 16 — THE FOREIGN SECTOR — MOHR 6E, §16.2, PP. 325–329 • MEDIUM

Spain and Greece produce olives and tobacco. With a given unit of resources, Spain can produce either 100 kg of olives or 150 kg of tobacco, while Greece can produce either 300 kg of olives or 300 kg of tobacco. What are the opportunity costs, and which country has the comparative advantage in each product?

Verified Solution: Spain has comparative advantage in tobacco (0.67 < 1 olive) and Greece in olives (1 < 1.5 tobacco).

$$\text{Spain: } 1 \text{ Tobacco} = \frac{100}{150} = 0.67 \text{ Olives}, \ 1 \text{ Olive} = \frac{150}{100} = 1.5 \text{ Tobacco}.$$
$$\text{Greece: } 1 \text{ Tobacco} = \frac{300}{300} = 1 \text{ Olive}, \ 1 \text{ Olive} = 1 \text{ Tobacco}.$$

First-Principles Proof: Correct! Spain sacrifices only 0.67 kg of olives per kg of tobacco (vs 1 kg in Greece), while Greece sacrifices only 1 kg of tobacco per kg of olives (vs 1.5 kg in Spain).

CHAPTER 16 — THE FOREIGN SECTOR — MOHR 6E, §16.2 & §16.4, PP. 326–336 • M

Australia can produce 150 surfboards or 900 bottles of sunscreen per month, whereas the U
1.70 AUS. $Which country has a relative advantage in surfboards, and what will A

Verified Solution: Australia has relative advantage in surfboards (costing 6 vs 8 bottles); 5

$$\text{Opp. Cost of 1 Surfboard:}$$

First-Principles Proof: Correct! Australia gives up 6 bottles of sunscreen per surfboard cor

CHAPTER 16 — THE FOREIGN SECTOR — MOHR 6E, §16.4, PP. 336–340 • EASY

Suppose the South African Rand appreciates against the US dollar (for example, moving from R19.00/1 $to R17.00/1) due to strong capital inflows and rising commodity demand. How will this appreciation impact South Africa's export prices (in foreign currency) and import prices (in Rand), ceteris paribus?

Verified Solution: South African export prices in foreign currency increase, while import prices in Rand decrease.

$$\text{Rand Appreciation } (E \downarrow) \implies P_{\text{exports in USD}} \uparrow \implies X \downarrow; \quad P_{\text{imports in ZAR}} \downarrow \implies Z \uparrow.$$

First-Principles Proof: Correct! A stronger Rand makes South African goods more expensive for foreign buyers in USD, while foreign goods become cheaper for South Africans in ZAR.

CHAPTER 16 — THE FOREIGN SECTOR — MOHR 6E, §16.2, PP. 325–329 • EASY

Suppose Japan and Portugal produce only Wagyu beef and Seafood. If all resources are ful
Compute the opportunity cost of producing 1 kg of Wagyu beef in both countries and ider

Verified Solution: Japan sacrifices 2.00 kg Seafood and Portugal sacrifices 1.36 kg, giving

$$\text{Japan: } 1 \text{ kg Wagyu} =$$

First-Principles Proof: Correct! Japan gives up 2000/1000 = 2.00 kg of Seafood per kg of

CHAPTER 16 — THE FOREIGN SECTOR — MOHR 6E, §16.2, PP. 325–329 • MEDIUM

Suppose 1 hour of labour in South Africa can produce either 6 metres of cloth or 4 cars, whereas 1 hour of labour in Zimbabwe can produce either 1 metre of cloth or 5 cars. Determine the opportunity cost of producing 1 metre of cloth and 1 car in each country, and state which country should specialise in each product.

Verified Solution: SA has comparative advantage in cloth (0.67 < 5 cars) and Zimbabwe has comparative advantage in cars (0.20 < 1.50m cloth).

$$\text{SA: } 1\text{m cloth} = \frac{4}{6} = 0.67 \text{ cars}, \ 1 \text{ car} = \frac{6}{4} = 1.50\text{m cloth}. \quad \text{Zim: } 1\text{m cloth} = \frac{5}{1} = 5 \text{ cars}, \ 1 \text{ car} = \frac{1}{5} = 0.20\text{m cloth}.$$

First-Principles Proof: Correct! SA sacrifices only 4/6 = 0.67 cars per metre of cloth (vs 5 cars in Zimbabwe), while Zimbabwe sacrifices only 1/5 = 0.20m of cloth per car (vs 1.50m in SA).

SIX INSTRUMENTS TO RESTRICT IMPORTS AND WELFARE EFFECTS OF AN IMPORT

List the six primary policy instruments governments use to restrict imports or influence inte
autarky equilibrium price.

Verified Solution: Instruments: Tariffs, quotas, domestic subsidies, non-tariff barriers, excl

$$\text{Six Instruments: (1) Import tariffs, (2) Import quotas, (3) Subsidies to d}$$
$$\text{Tariff Impact: Domestic price rises from } P_w \text{ to } P$$
$$\text{Welfare: Domestic producers gain surplus, government coll}$$

First-Principles Proof: Correct! All six trade policy instruments are accurately identified, an

OPPORTUNITY COST OF MAIZE VS STEEL BETWEEN TWO TRADING PARTNERS • MEDIUM

Using identical resource endowments, Country A can produce either 80 tons of Maize or 40 tons of Steel, whereas Country B can produce either 30 tons of Maize or 30 tons of Steel. Determine (1) which country holds the absolute advantage in both goods, (2) each country's opportunity cost of producing 1 ton of Steel and 1 ton of Maize, and (3) the pattern of comparative advantage.

Verified Solution: Country A has absolute advantage in both goods; Country A has comparative advantage in Maize (cost = 0.5t Steel), and Country B has comparative advantage in Steel (cost = 1.0t Maize).

$$\text{Absolute Advantage: Country A produces more Maize } (80 > 30) \text{ and more Steel } (40 > 30)$$
$$\text{Opportunity Cost of 1 ton of Steel: Country A} = \frac{80}{40} = 2.0 \text{ tons Maize}; \quad \text{Country B} = \frac{30}{30} = 1.0 \text{ ton Maize}$$
$$\text{Opportunity Cost of 1 ton of Maize: Country A} = \frac{40}{80} = 0.5 \text{ tons Steel}; \quad \text{Country B} = \frac{30}{30} = 1.0 \text{ ton Steel}$$
$$\text{Comparative Advantage: Country A in Maize } (0.5 < 1.0 \text{ Steel}); \quad \text{Country B in Steel } (1.0 < 2.0 \text{ Maize})$$

First-Principles Proof: Correct! Even though Country A is more productive in both goods, Country B sacrifices only 1 ton of Maize per ton of Steel (vs 2 tons in Country A), giving B comparative advantage in Steel.

CHAPTER 17 Mohr 6e, pp. 347–376 (PDF pp. 823–875) • Simple Keynesian Model

Chapter 17: A Simple Keynesian Model of the Economy

Core assumptions of the simple closed no-government Keynesian model, Say's Law vs Keynes's Law, the consumption function and its three characteristics, saving function, autonomous investment, the 45° line, equilibrium income, and the simple multiplier.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Consumption & Saving Functions

$$C = \bar{C} + cY, \quad S = -\bar{C} + (1-c)Y, \quad c + s = 1$$

Meaning: Autonomous consumption C_bar (>0) is independent of income; induced consumption cY depends on the marginal propensity to consume c (0 < c < 1).

Simple Closed-Economy Equilibrium

$$Y_0 = \alpha \bar{A} = \left(\frac{1}{1-c}\right)(\bar{C} + \bar{I})$$

Meaning: Equilibrium occurs along the 45° line where Aggregate Spending A = C + I equals Total Production/Income Y.

The Simple Multiplier Process

$$\alpha = \frac{\Delta Y}{\Delta \bar{A}} = \frac{1}{1-c} = \frac{1}{s} > 1$$

Meaning: An autonomous injection (e.g. ΔI) triggers a geometric chain of induced consumption rounds, expanding equilibrium income by a multiple of the initial spending change.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

Assumptions of the Simple Keynesian Model & Say's Law (Mohr §17.2, pp. 351–352)

The simple model assumes: (1) The economy consists only of households and firms (closed, no government), (2) Prices and wages are fixed, (3) The money market and interest rates are constant, and (4) Aggregate spending (demand) determines aggregate production (income). Unlike Classical Say's Law ('supply creates its own demand', Y → A), Keynes showed that aggregate spending drives output (A → Y) and equilibrium can persist below full employment.

Worked Proof: At any output level below equilibrium (to the left of the 45° intersection), Aggregate Spending exceeds Production (A > Y), causing unplanned inventory depletion and prompting firms to expand output.

Three Characteristics of the Consumption Function (Mohr §17.3, pp. 353–356)

Keynes's consumption function C = C_bar + cY has three fundamental properties: (1) Consumption is strictly positive even when income is zero (C_bar > 0, financed by dissaving/past wealth), (2) Consumption increases as income increases (positive slope, c > 0), and (3) When income increases, consumption increases by LESS than the increase in income (0 < c < 1, because households save part of every extra Rand).

Worked Proof: If C = 100 + 0.8Y, then at Y = 0 households spend R100 (dissaving S = -100). For every R100 rise in Y, consumption rises by R80 (MPC = 0.8) and saving rises by R20 (MPS = 0.2).

The Multiplier & Unplanned Inventory Adjustment (Mohr §17.8, pp. 366–371)

When investment increases by ΔI, total spending jumps vertically by ΔI. Because each Rand spent becomes income for someone else who spends a fraction c of it, total income expands by ΔY = [1 + c + c^2 + c^3 + ...] ΔI = [1/(1-c)] ΔI. Consequently, the horizontal increase in equilibrium income (ΔY) is strictly greater than the vertical increase in investment (ΔI).

Worked Proof: With MPC c = 0.75, the multiplier is α = 1/(1 - 0.75) = 4. A R50m increase in autonomous investment raises equilibrium income by 4 × R50m = R200m.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Assumptions of the Simple Keynesian Model & Say's Law): Believing that Keynesian equilibrium (A = Y) automatically guarantees full employment of all labour.

First-Principles Law: Macroeconomic equilibrium in the Keynesian model simply means planned spending equals production (A = Y); this equilibrium income Y0 can be far below full-employment output Yf.

Fallacy (Three Characteristics of the Consumption Function): Confusing a movement along the consumption curve (caused by a change in income Y) with a vertical shift of the consumption curve (caused by non-income factors like wealth or interest rates).

First-Principles Law: Income Y is on the horizontal axis, so ΔY causes a movement along C = C_bar + cY (induced consumption), whereas a change in autonomous consumption ΔC_bar shifts the entire intercept vertically.

Fallacy (The Multiplier & Unplanned Inventory Adjustment): Thinking that higher household thrift (an autonomous rise in the saving rate) always increases total national saving in the simple Keynesian model.

First-Principles Law: By the Paradox of Thrift (Box 17-6), an autonomous attempt to save more reduces C_bar and lowers the multiplier, contracting equilibrium income Y0 and leaving total saving unchanged at S = I.

STANDALONE ANALYTICAL PROBLEMS & DERIVATIONS (5 PROBLEMS)

CHAPTER 17 — A SIMPLE KEYNESIAN MODEL OF THE ECONOMY — MOHR 6E, §17.1–17.6, PP. 348–361 • EASY

Which one of the following statements accurately describes the simple Keynesian model of income determination and its relationship to the 45-degree line?

Verified Solution: Any point on the aggregate spending line above the 45° line means spending exceeds production (A > Y).

$$\text{Keynes's Law: } A \to Y. \quad \text{Above } 45^\circ \text{ line } (Y < Y_0) : A > Y \implies \Delta\text{Inventories} < 0 \implies Y \uparrow.$$

First-Principles Proof: Correct! Above the 45° line (to the left of equilibrium), aggregate spending A exceeds output Y, causing unplanned inventory depletion and an expansion in output.

CHAPTER 17 — A SIMPL

In a simple closed Keynesi
(c) is 0.75, and autonomou
equilibrium level of incom

Verified Solution: Equili

$$\alpha = \frac{1}{1 - 0.75} = 4$$

First-Principles Proof: Co
consumption rounds that

CHAPTER 17 — A SIMPLE KEYNESIAN MODEL OF THE ECONOMY — MOHR 6E, §17.4–17.7, PP. 361–376 • MEDIUM

In a simple closed economy without a government, the saving function is given by S = -120 + 0.25Y and autonomous investment is I = 80. Determine the consumption function C(Y), the simple multiplier (α), equilibrium income (Y0), and what happens if households attempt to save more by shifting autonomous saving upward (the Paradox of Thrift).

Verified Solution: C = 120 + 0.75Y, α = 4.0, Y0 = 800; higher autonomous saving reduces equilibrium income while leaving total S = I = 80.

$$S = -\bar{C} + (1-c)Y = -120 + 0.25Y \implies \bar{C} = 120,\ c = 0.75. \quad \alpha = \frac{1}{0.25} = 4.0. \quad Y_0 = 4(120 + 80) = 800.$$

First-Principles Proof: Correct! Since S = -C̄ + (1-c)Y, C = 120 + 0.75Y, α = 1/0.25 = 4, Y0 = 4(200) = 800, and the Paradox of Thrift lowers Y0 without raising equilibrium saving (which remains equal to I = 80).

CHAPTER 17 — A SIMPL

Compare Classical (Say's L
(A): which statement accur

Verified Solution: Say's
Y).

$$\text{Say's Law: } Y \implies$$

First-Principles Proof: Co
Keynesian model aggregat

CHARACTERISTICS OF THE CONSUMPTION FUNCTION, INCOME VS DISPOSABLE INCOME, AND INJECTIONS/LEAKAGES • EASY

In the Keynesian macroeconomic model: (1) What are the three fundamental characteristics of the household consumption function $C = \bar{C} + cY_d$? (2) How does total National Income (Y) differ from Disposable Income (Y_d)? (3) How do Investment (I) and Saving (S) classify as injections or leakages, and what are their effects on the multiplier α and equilibrium income Y_e?

Verified Solution: (1) C > 0 at Yd = 0, C rises with Yd, and 0 < c < 1; (2) Disposable income is after-tax income (Yd = Y - T); (3) Investment is an injection raising Ye, while Saving is a leakage.

Three Characteristics: (1) $C > 0$ even when $Y_d = 0$ (autonomous consumption $\bar{C} > 0$); (2) C increases as Y_d increases ($c > 0$); (3) $\Delta C < \Delta Y_d$ because $0 < c < 1$

Y vs Y_d : Y is total factor income earned across the economy, whereas $Y_d = Y - T = (1-t)Y$ is after-tax income available to households for consumption or saving

I vs S : I is an autonomous injection (raises Y_e via ΔA, leaves α unchanged); S is a leakage (higher saving propensity $s = 1 - c$ lowers c, decreasing both α and Y_e)

First-Principles Proof: Correct! These capture the three Keynesian consumption axioms, the tax wedge between Y and Yd, and the exact roles of I (autonomous injection) and S (income-induced leakage).

CHAPTER 18 Mohr 6e, pp. 377–405 (PDF pp. 876–930) • Open Keynesian Model

Chapter 18: The Keynesian Model with a Government and a Foreign Sector

Introducing government spending (G) and proportional taxes (T = tY), disposable income, open-economy exports (X) and induced imports (Z = Z_bar + mY), leakages vs injections, open-economy multiplier, and fiscal policy adjustments.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Consumption with Proportional Income Tax

$$T = tY, \quad Y_d = (1-t)Y \implies C = \bar{C} + c(1-t)Y$$

Meaning: Proportional taxes t reduce disposable income Yd, flattening the slope of the consumption schedule from c to c(1 - t).

Multiplier with Government Sector (Closed)

$$\alpha_G = \frac{1}{1 - c(1-t)}$$

Meaning: Government spending G is autonomous and does NOT change the multiplier, whereas the tax rate t is a leakage that reduces the multiplier.

Open-Economy Multiplier (with Induced Imports)

$$\alpha_{\text{open}} = \frac{1}{1 - c(1-t) + m}, \quad \bar{A} = \bar{C} + \bar{I} + \bar{G} + \bar{X} - \bar{Z}$$

Meaning: Both proportional taxes (t) and the marginal propensity to import (m) act as leakages from the domestic spending stream, reducing the size of the multiplier.

Fiscal Policy Gap to Full Employment

$$\Delta\bar{G} = \frac{Y_f - Y_0}{\alpha}$$

Meaning: To close a recessionary output gap (Yf - Y0), government spending must increase by the income gap divided by the multiplier.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

How G and t Affect the Aggregate Spending Curve Differently (Mohr §18.1, pp. 378–386)

Government spending G is an autonomous injection: it shifts the Aggregate Spending curve upward in parallel by ΔG without changing its slope or the multiplier. Proportional taxes T = tY are an induced leakage: an increase in the tax rate t rotates (pivots) the Aggregate Spending curve downward, reducing its slope c(1-t) and shrinking the multiplier.

Worked Proof: If G rises by R4m while t is simultaneously raised from 0.167 to 0.25, the intercept A_bar rises from R65m to R69m, but the multiplier falls from 1.60 to 1.48, causing equilibrium income to fall from R104m to R102.22m.

Exports, Induced Imports & the Open-Economy Multiplier (Mohr §18.2–18.3, pp. 387–393)

Exports X depend on foreign income and exchange rates, so X is an autonomous injection (+X_bar). Imports Z = Z_bar + mY consist of autonomous imports (Z_bar, subtracted from A_bar) and induced imports (mY, which rise as domestic income rises, reducing the slope of net spending to c(1-t) - m).

Worked Proof: Given C_bar = 150, I = 300, G = 400, X = 250, Z = 50 + 0.15Y, c = 0.75, t = 0.25: A_bar = 150 + 300 + 400 + 250 - 50 = 1050; α = 1/[1 - 0.75(0.75) + 0.15] = 1/0.5875 = 1.70; Y0 = 1.70 × 1050 = 1785 (or exact 1787.23).

Injections vs. Leakages Table (Core Synthesis) (Mohr Appendix 18-1, p. 402)

In the open circular flow with government: Investment (I), Government Spending (G), and Exports (X) are INJECTIONS that increase equilibrium income Y0 without changing α. Saving (s), Taxation (t), and Imports (m) are LEAKAGES (withdrawals): an increase in s, t, or m decreases the multiplier α and decreases equilibrium income Y0.

Worked Proof: Variable G is an Injection, α remains unchanged, and Ye increases; Variable t is a Leakage, α decreases, and Ye decreases.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (How G and t Affect the Aggregate Spending Curve Differently): Saying that an increase in government spending G increases the size of the multiplier α.

First-Principles Law: The multiplier α = 1/[1 - c(1-t) + m] depends ONLY on the marginal propensities (c, m) and the tax rate (t). Autonomous G changes the intercept A_bar, never α.

Fallacy (Exports, Induced Imports & the Open-Economy Multiplier): Adding autonomous imports Z_bar (+50) into autonomous spending A_bar instead of subtracting it (-50).

First-Principles Law: Imports represent domestic spending leaking to foreign producers (X - Z = X_bar - Z_bar - mY), so autonomous imports Z_bar must always be subtracted from A_bar.

Fallacy (Injections vs. Leakages Table (Core Synthesis)): Claiming that an increase in the marginal propensity to consume (c) reduces the multiplier.

First-Principles Law: Because households re-spend a larger fraction of income domestically when c rises, α = 1/[1 - c(1-t) + m] increases and equilibrium income Y0 increases.

STANDALONE ANALYTICAL PROBLEMS & DERIVATIONS (6 PROBLEMS)

CHAPTER 18 — THE KEYNESIAN MODEL WITH A GOVERNMENT AND A FOREIGN SECTOR — MOHR 6E, §18.2–18.3, PP. 387–393 • HARD

Suppose the following equations describe the open economy of Portofino, Italy: Autonomous consumption = 100, MPC (c) = 0.8, proportional tax rate (t) = 0.2 (so T = 0.2Y), Investment (I) = 100, Government spending (G) = 250, Exports (X) = 300, and Imports Z = 20 + 0.1Y. Calculate the open-economy multiplier (rounded to 2 decimal places) and the equilibrium level of income.

Verified Solution: The multiplier is 2.17 and equilibrium income is 1 584.10 (or 1 586.96 unrounded) with Ā = 730.

$$\bar{A} = 100 + 100 + 250 + 300 - 20 = 730.$$
$$\alpha = \frac{1}{1 - c(1-t) + m} = \frac{1}{1 - 0.8(1-0.2) + 0.1} = \frac{1}{1 - 0.64 + 0.1} = \frac{1}{0.46} \approx 2.17.$$
$$Y_0 = 2.17 \times 730 = 1\,584.10 \text{ (or } 730/0.46 = 1\,586.96).$$

First-Principles Proof: Correct! Autonomous spending is 100 + 100 + 250 + 300 - 20 = 730, and α = 1/[1 - 0.8(0.8) + 0.1] = 1/0.46 = 2.17, giving Y0 = 1 584.10 (or 1 586.96 exact).

CHAPTER 18 — THE KEYNESIAN MODEL WITH A GOVERNMENT AND A FOREIGN SECTOR — M

Consider an open economy with government where: Government expenditure (G) = R400, Exports (X) = R consumption (C̄) = R150, Investment (I) = R300, Full-employment output (Yf) = R1 900, MPC (c) = 0.75, a (rounded to 2 decimals), equilibrium income, and the new equilibrium income if investment falls to R250.

Verified Solution: Multiplier is 1.70, equilibrium income is R1 785 (R1 787.23 exact), falling to R1 700 w

$$\bar{A} = 150 + 300 + 400 + 250 - 50 = 1\,050.$$
$$\alpha = \frac{1}{1 - 0.75(1-0.25) + 0.15} = \frac{1}{1 - 0.5625 + 0.15} = \frac{1}{0.5}$$
$$Y_0 = 1.70 \times 1\,050 = 1\,785 \text{ (exact } 1\,787.23). \quad \Delta I = -50 \implies \Delta Y =$$

First-Principles Proof: Correct! α = 1/0.5875 = 1.70 and Ā = 1 050, so Y0 = 1.70 × 1 050 = R1 785 (or R1 1.70 = R85 to R1 700.

CHAPTER 18 — THE KEYNESIAN MODEL WITH A GOVERNMENT AND A FOREIGN SECTOR — MOHR 6E, §18.4, PP. 393–398 • HARD

In the South African Keynesian model (in R millions): C̄ = 30, Ī = 10, Ḡ = 20, X̄ = 10, Z̄ = 5, MPC (c) = 0.6, average tax rate (t) = 0.167, and MPM (m) = 0.125. Initial equilibrium income is R104m (with α = 1.60 and Ā = 65). Suppose the government introduces a basic income grant that raises G by R4m (to R24m) and finances it by raising the tax rate t to 0.25. What are the new equilibrium income and tax revenue?

Verified Solution: New equilibrium income is R102.22 million and government tax revenue is R25.56 million.

$$\bar{A}_{\text{new}} = 30 + 10 + 24 + 10 - 5 = 69. \quad c(1 - t_{\text{new}}) = 0.6(0.75) = 0.45.$$
$$Y_{\text{new}} = \frac{69}{1 - 0.45 + 0.125} = \frac{69}{0.675} = \text{R}102.22\text{m}. \quad T_{\text{new}} = 0.25 \times 102.22 = \text{R}25.56\text{m}.$$

First-Principles Proof: Correct! With Ā_new = 69 and the new denominator 1 - 0.6(0.75) + 0.125 = 0.675, Y_new = 69 / 0.675 = R102.22m and T = 0.25 × 102.22 = R25.56m.

CHAPTER 18 — THE KEYNESIAN MODEL WITH A GOVERNMENT AND A FOREIGN SECTOR — M

You are given the following model for the Italian economy: Consumption C = 85 + 0.5Yd; Investment I = 0.25Y; Disposable income Yd = Y - T; Exports X = 30; Imports Z = 30 + 0.3Yd. Calculate equilibrium incom government budget balance (T - G), and household savings (S).

Verified Solution: Equilibrium Y is 270.59, Taxes are 67.65, Budget Deficit is -2.35, and Savings equal 16

$$\bar{A} = 85 + 75 + 70 + 30 - 30 = 230. \quad Y = 230 + (0.5 - 0.3)(1 - 0.25)Y = 230 -$$
$$T = 0.25(270.59) = 67.65; \quad \text{Budget} = 67.65 - 70 = -2.35; \quad S = -85 +$$

First-Principles Proof: Correct! With Ā = 230 and induced slope (0.5 - 0.3)(0.75) = 0.15, Y = 230 / 0.85 = (deficit), and S = 16.47.

CHAPTER 18 — THE KEYNESIAN MODEL WITH A GOVERNMENT AND A FOREIGN SECTOR — MOHR 6E, §18.1, PP. 381–387 • EASY

In the Keynesian model with a government sector, compare how an increase in government spending (G) versus an increase in the proportional tax rate (t) affects (i) the circular flow (injection vs leakage), (ii) the size of the multiplier (α), and (iii) the equilibrium level of income (Ye).

Verified Solution: G is an injection that leaves α unchanged and raises Ye; t is a leakage that reduces α (flattens A) and lowers Ye.

$$G: \text{Autonomous Injection } (\alpha \text{ unchanged}, \ Y_e \uparrow). \quad t: \text{Induced Leakage } \left(\alpha = \frac{1}{1 - c(1-t) + m} \downarrow, \ Y_e \downarrow\right).$$

First-Principles Proof: Correct! Government spending G is purely autonomous (shifting A parallel upward without changing α), whereas the tax rate t reduces the slope c(1-t) and lowers both α and Ye.

EQUILIBRIUM OUTPUT AND INVESTMENT SHOCK PROPAGATION IN OPEN ECONOMIES • HAR

Perform two open-economy Keynesian analyses: (Model 1) An economy has $\bar{C} = 100, c = 0.8, t = 0.$ $0.1Y$, yielding $\alpha = 2.17$ $(1/0.46)$ and initial equilibrium $Y_0 = 1,584.10$ (or $1,586.96$). If autonomou the new equilibrium income Y_1? (Model 2) A second economy has $\bar{C} = 400, c = 0.6, t = 0.25, \bar{I} = 4$ $0.15Y$. Calculate its multiplier α and equilibrium income Y_e.

Verified Solution: Model 1: ΔY = 325.50 and Y1 = 1,909.60 (unrounded 1,913.04); Model 2: Ā = 1,200, c

$$\text{Model 1 Shock: } \Delta Y = \alpha \times \Delta I = 2.17 \times 150 = 325.50 \implies Y_1 = 1,584.10 + 325.5($$
$$\text{Model 2 Autonomous Spending: } \bar{A} = 400 + 400 + 300 + 200$$
$$\text{Model 2 Multiplier: } c(1-t) - m = 0.6(1 - 0.25) - 0.15 = 0.45 - 0.15 = 0.30$$
$$\text{Model 2 Equilibrium: } Y_e = \frac{1,200}{0.70} = 1,714.29 \text{ (or } 1.43 \times 1,2$$

First-Principles Proof: Correct! In Model 1, the +150 investment shock multiplies by 2.17 to add 325.50; = 1.43 and Ye = 1,200/0.70 = 1,714.29.

CHAPTER 19 Mohr 6e, pp. 407–436 (PDF pp. 931–982) • AD–AS & Policy

Chapter 19: More on Macroeconomic Theory and Policy

The Aggregate Demand–Aggregate Supply (AD–AS) model, variable price levels, why AD slopes downward (wealth, interest rate, trade effects), SRAS vs LRAS, monetary transmission mechanism channels, four policy lags, and Monetarist vs Keynesian schools.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Aggregate Demand Downward Slope Effects

$$P \uparrow \implies \begin{cases} (M/P) \downarrow \implies C \downarrow & \text{(Wealth/Real Balance)} \\ i \uparrow \implies I \downarrow & \text{(Interest Rate Effect)} \\ (X - Z) \downarrow & \text{(International Trade)} \end{cases}$$

Meaning: A rise in the domestic price level P reduces real money balances, raises interest rates, and erodes export competitiveness, reducing real output demanded.

Monetary Transmission Mechanism

$$\Delta i_{\text{repo}} \rightarrow \Delta i_{\text{market}} \rightarrow \Delta I \text{ (and } \Delta C, \Delta E, \Delta P_{\text{assets}}) \rightarrow \Delta \text{AD} \rightarrow \Delta P, \Delta Y$$

Meaning: Links monetary policy changes in the repo rate to real economic output and inflation via interest rate, exchange rate, asset price, and credit channels.

Equation of Exchange (Quantity Theory of Money)

$$M \times V = P \times Y \implies \text{If } V, Y \text{ constant}, \%\Delta P = \%\Delta M$$

Meaning: Monetarists argue that velocity V is stable and output Y is near full capacity, so excessive money growth directly causes inflation.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

The AD–AS Framework: Relaxing the Fixed-Price Assumption (Mohr §19.1, pp. 408–417)

While the Keynesian model of Ch 17–18 assumes fixed prices, the AD–AS model places the general price level (P) on the vertical axis and real output (Y) on the horizontal axis. Short-run AS (SRAS) slopes upward due to sticky nominal wages and rising marginal costs; Long-run AS (LRAS) is vertical at full-employment potential output Yf because in the long run output depends solely on supply-side productive capacity.

Worked Proof: An expansionary fiscal boost (↑G) shifts AD rightward along an upward-sloping SRAS, increasing real output Y (reducing unemployment) at the cost of a higher price level P.

Adverse Supply Shocks & Stagflation in AD–AS (Mohr §19.1, pp. 415–417)

An increase in unit production costs (e.g. an oil price spike, Eskom electricity tariff surge, or wage hikes exceeding productivity growth) shifts the SRAS curve upward/leftward. The economy experiences STAGFLATION: a simultaneous rise in the price level (↑P, cost-push inflation) and contraction in real output and employment (↓Y).

Worked Proof: If policymakers respond to a leftward SRAS shift solely by expanding AD (cutting interest rates), they restore output Y0 only by worsening inflation even further; if they contract AD, they tame inflation only by deepening the recession.

Four Policy Lags & Monetarist vs. Keynesian Views (Mohr §19.3–19.4, pp. 423–432)

Stabilisation policy faces four chronological lags: (1) Recognition lag (time to collect data confirming a shock, same for both policies), (2) Decision lag (very long for fiscal policy due to annual parliamentary budget process; very short for SARB MPC), (3) Implementation lag (short for monetary policy), and (4) Impact/outside lag (long for monetary policy as investment responds slowly; shorter for direct fiscal spending).

Worked Proof: Monetarists (Milton Friedman) view the market economy as inherently stable and warn that long, variable policy lags make activist Keynesian fine-tuning destabilising.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (The AD–AS Framework: Relaxing the Fixed-Price Assumption): Explaining the downward slope of the macroeconomic AD curve using the microeconomic substitution effect between two individual goods.

First-Principles Law: When the aggregate price level P rises, all domestic goods become more expensive together; AD slopes downward due to the real balance (wealth) effect, the Keynes interest rate effect, and the international trade effect.

Fallacy (Adverse Supply Shocks & Stagflation in AD–AS): Thinking demand-management policies (monetary or fiscal policy alone) can easily cure stagflation.

First-Principles Law: Because monetary and fiscal policies shift AD rather than AS, they cannot simultaneously reverse both higher prices and lower output after an adverse supply shock.

Fallacy (Four Policy Lags & Monetarist vs. Keynesian Views): Claiming that monetary policy has a longer decision lag than fiscal policy.

First-Principles Law: Fiscal policy has a long decision/inside lag (parliamentary approval) but a relatively short impact lag, whereas monetary policy has a very short decision lag (SARB MPC meets every 2 months) but a long impact/outside lag (12–24 months).

STANDALONE ANALYTICAL PROBLEMS & DERIVATIONS (4 PROBLEMS)

CHAPTER 19 — MORE ON MACROECONOMIC THEORY AND POLICY — MOHR 6E, §19.1–19.4, PP. 408–431 • MEDIUM

How do the views of Monetarist economists differ from Keynesian economists regarding economic fluctuations, and how do the decision lag and impact lag compare between fiscal policy and monetary policy?

Verified Solution: Keynesians see instability as endogenous; fiscal policy has a longer decision lag but shorter impact lag.

$$\text{Keynesian: Endogenous Instability} \implies \text{Active Policy.} \quad \text{Decision Lag: Fiscal} > \text{Monetary;} \quad \text{Impact Lag: Monetary} > \text{Fiscal.}$$

First-Principles Proof: Correct! Keynesians view fluctuations as inherent to market economies, and parliamentary budgets give fiscal policy a long decision lag while repo changes have a long impact lag.

CHAPTER 19 — MORE ON MACROECONOMIC THEORY AND

In the Aggregate Demand–Aggregate Supply (AD–AS) model, whi slopes downward from left to right as the general price level (P) fa

Verified Solution: The wealth (real balance) effect, the interest r

$$P \downarrow \implies (1) \ (M/P) \uparrow \implies C \uparrow \ \text{(Wealth)}$$

First-Principles Proof: Correct! A lower price level raises the real improves net export competitiveness (boosting X - Z).

CHAPTER 19 — MORE ON MACROECONOMIC THEORY AND POLICY — MOHR 6E, §19.2–19.3, PP. 421–428 • MEDIUM

In macroeconomic policy debates (Chapter 19), how do Monetarists and Keynesians differ regarding the stability of the private sector, and how do the implementation and impact lags compare between fiscal policy and monetary policy?

Verified Solution: Monetarists view the private economy as stable; fiscal policy has a longer decision lag while monetary policy has a longer impact lag.

$$\text{Monetarists: Private sector inherently stable; avoid activist fine-tuning.} \quad \text{Fiscal: Long decision/implementation lag, short impact lag.}$$

First-Principles Proof: Correct! Monetarists believe markets are self-correcting and activist policy is destabilising; fiscal changes require parliamentary budget approval (long decision lag), while repo rate changes take 12–24 months to work (long impact lag).

CHAPTER 19 — MORE ON MACROECONOMIC THEORY AND

Trace the exchange-rate channel and asset-price (wealth) channel Committee (MPC) cuts the repo rate by 100 basis points:

Verified Solution: Lower interest rates trigger capital outflows t

$$\Delta i < 0 \implies \text{Capital outflows} \implies \text{Rand depreciat}$$

First-Principles Proof: Correct! Lower domestic yields reduce for and discounted cash flows raise asset wealth and consumption.

CHAPTER 20 Mohr 6e, pp. 437–466 (PDF pp. 983–1034) • Inflation & Phillips

Chapter 20: Inflation

Definition and measurement of inflation (CPI, PPI, Implicit GDP Deflator), distributional/economic/social effects, bracket creep, demand-pull vs cost-push inflation, structuralist & conflict approaches, SARB inflation targeting, and the Phillips curve.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

CPI Inflation Rate

$$\pi_t = \left(\frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}}\right) \times 100\%$$

Meaning: Percentage change in the cost of a fixed representative consumer basket (measured month-on-same-month of previous year or annual average on annual average).

GDP Deflator Inflation Rate

$$\text{Deflator}_t = \frac{\text{Nom GDP}_t}{\text{Real GDP}_t} \times 100, \quad \pi_t^{\text{Defl}} = \left(\frac{\text{Deflator}_t - \text{Deflator}_{t-1}}{\text{Deflator}_{t-1}}\right) \times 100\%$$

Meaning: Measures economy-wide inflation across all domestically produced final goods and services using a variable current-production basket.

Real Purchasing Power & Real Interest Rate

$$r_{\text{real}} \approx i_{\text{nominal}} - \pi$$

Meaning: When inflation π exceeds the nominal interest rate i, the real interest rate is negative, redistributing real wealth from creditors (savers) to debtors (borrowers).

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

CPI vs. PPI vs. Implicit GDP Deflator (Comparative Matrix) (Mohr §20.2, pp. 438–442)

Three price indicators differ in scope and basket structure: (1) CPI measures the cost of living for urban households using a fixed consumer basket that INCLUDES imported consumer goods and VAT; (2) PPI measures prices at the level of first significant commercial transaction (production/factory gate), EXCLUDES services and VAT, and includes capital & intermediate goods; (3) The GDP Deflator covers ALL domestically produced final goods/services (including exports & capital goods) but EXCLUDES all imports.

Worked Proof: CPI measures cost of living vs PPI measures cost of production; CPI includes services & VAT vs PPI excludes services & VAT; CPI basket covers consumer goods vs PPI includes capital and intermediate goods.

Distributional Effects of Inflation & Bracket Creep (Mohr §20.3, pp. 442–446)

Inflation redistributes income and wealth arbitrarily: (1) Debtors (borrowers) gain at the expense of creditors (lenders) because loans are repaid in Rand with lower real purchasing power; (2) Young borrowers with mortgages gain while elderly pensioners on fixed nominal annuities lose; (3) The government always gains via 'bracket creep' (fiscal drag) as nominal wage rises push taxpayers into higher progressive SARS tax brackets without any increase in real income, and because the state is a major net debtor.

Worked Proof: If a worker's salary rises by 7% to match 7% inflation, their real pre-tax income is unchanged, yet under an unindexed progressive tax table their effective tax rate rises, reducing their real disposable income.

Demand-Pull vs. Cost-Push Inflation & the Phillips Curve (Mohr §20.4–20.6, pp. 446–463)

Demand-pull inflation occurs when aggregate demand (C + I + G + X) outstrips productive capacity, shifting AD rightward (↑P and ↑Y). Cost-push inflation is triggered by 5 sources: (1) Wage increases exceeding productivity, (2) Higher prices of imported capital/intermediate goods (e.g. oil or weak Rand), (3) Increased profit margins, (4) Decreased labour productivity, and (5) Natural disasters, shifting AS leftward (↑P and ↓Y). The short-run Phillips curve illustrates an inverse trade-off between inflation (π) and unemployment (U); stagflation shifts the Phillips curve rightward.

Worked Proof: To combat demand-pull inflation, SARB applies contractionary monetary policy (raising the repo rate) and Treasury applies contractionary fiscal policy (cutting G or raising taxes T), shifting AD leftward.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (CPI vs. PPI vs. Implicit GDP Deflator (Comparative Matrix)): Calling a one-off jump in the price of bread or petrol 'inflation'.

First-Principles Law: Inflation is a CONTINUOUS and CONSIDERABLE rise in prices IN GENERAL—not a once-off price level shock or a relative price change in a single commodity.

Fallacy (Distributional Effects of Inflation & Bracket Creep): Assuming that everyone loses equally during inflation.

First-Principles Law: Inflation redistributes real wealth from lenders/fixed-income holders to borrowers and the government (via bracket creep and erosion of real public debt).

Fallacy (Demand-Pull vs. Cost-Push Inflation & the Phillips Curve): Using contractionary monetary/fiscal policy aggressively against pure cost-push inflation without side effects.

First-Principles Law: Restricting AD when inflation is cost-push curbs price rises only by worsening the contraction in real output Y and increasing unemployment further.

STANDALONE ANALYTICAL PROBLEMS & DERIVATIONS (7 PROBLEMS)

CHAPTER 20 — INFLATION — MOHR 6E, §20.2, PP. 438–442 • MEDIUM

The macroeconomic table below shows Nominal GDP and Real GDP (in R millions) for two consecutive years: in Year 1 (Period/Period 2), Nominal GDP = 5 130 703 and Real GDP = 4 028 475; in Year 2 (Period/Period 2), Nominal GDP = 5 643 773 and Real GDP = 4 391 038. Calculate the implicit GDP deflator for both years and the resulting annual inflation rate (rounded to 2 decimal places).

Verified Solution: The GDP deflator rises from 127.36 to 128.53, yielding an annual inflation rate of 0.92%.

$$\text{Deflator}_1 = \frac{5\,130\,703}{4\,028\,475} \times 100 = 127.36; \quad \text{Deflator}_2 = \frac{5\,643\,773}{4\,391\,038} \times 100 = 128.53.$$
$$\pi = \frac{128.53 - 127.36}{127.36} \times 100 = 0.92\%.$$

First-Principles Proof: Correct! Deflator Year 1 = (5 130 703 / 4 028 475) × 100 = 127.36; Deflator Year 2 = (5 643 773 / 4 391 038) × 100 = 128.53; inflation = [(128.53 - 127.36)/127.36] × 100 = 0.92%.

CHAPTER 20 — INFLATION — MOHR 6E, §13.2 & §20.2, PP. 243, 4

An economy produces three goods (Period is the base year): Pizza (Perio Cream (Period 1: Q=15, P=R8.00; Period 2: Q=22.5, P=R8.80). Calculate P rate.

Verified Solution: Period GDP is R860; Period 2 Nominal is R1 419 and

$$\text{GDP}_1^{\text{Nom}} =$$
$$\text{GDP}_2^{\text{Nom}} = 36(24.75) + 75(4.4) + 2$$
$$\text{Deflator}$$

First-Principles Proof: Correct! Period GDP = 540 + 200 + 120 = R860; inflation = 10%.

CHAPTER 20 — INFLATION — MOHR 6E, §20.2, PP. 438–442 • EASY

Which statement accurately contrasts the Consumer Price Index (CPI), the Producer Price Index (PPI), and the Implicit GDP Deflator in South Africa?

Verified Solution: CPI includes services and imported consumer goods; PPI excludes services and includes capital goods.

\text{CPI: Urban consumer basket (incl. imports & services)}. \quad \text{PPI: Production gate (excl. services & VAT, incl. capital/intermediate)}. \quad \text{Deflator: All domestic final output (excl. imports)}.

First-Principles Proof: Correct! CPI measures household cost of living (including services, VAT, and imported consumer goods), whereas PPI measures producer prices (excluding services and VAT, including capital/intermediate goods).

CHAPTER 20 — INFLATION — MOHR 6E, §20.2–20.3, PP. 438–445

A household spent R4 200 on Food, R1 900 on Transport, R5 800 on Ren household expenditure increase to maintain the exact same real consum

Verified Solution: Expenditure rises by R897.60 (to R14 097.60); CPI inc

$$\text{Initial Total} = 4\,200 + 1\,900 + 5\,800 + 1$$

First-Principles Proof: Correct! 6.8% of R13 200 is R897.60 (new total R domestic production.

CHAPTER 20 — INFLATION — MOHR 6E, §20.2, PP. 441–443 • MEDIUM

According to central bank national accounting statistics, Nominal GDP was R2 917 539 million in Period 1 and R3 155 195 million in Period 2, while Real GDP (at constant base prices) was R1 905 736 million in Period 1 and R1 954 303 million in Period 2. Calculate the implicit GDP Deflator for Period 1 and Period 2 and the resulting annual inflation rate.

Verified Solution: The Period 1 GDP Deflator is 153.09, the Period 2 GDP Deflator is 161.45, and the inflation rate is 5.46%.

$$\text{Defl}_1 = \frac{2\,917\,539}{1\,905\,736} \times 100 = 153.09; \quad \text{Defl}_2 = \frac{3\,155\,195}{1\,954\,303} \times 100 = 161.45.$$
$$\pi_2 = \frac{161.45 - 153.09}{153.09} \times 100 = 5.46\%.$$

First-Principles Proof: Correct! Deflator 1 = (2 917 539 / 1 905 736) × 100 = 153.09 and Deflator 2 = (3 155 195 / 1 954 303) × 100 = 161.45; percentage change = (161.45 - 153.09) / 153.09 × 100 = 5.46%.

CALCULATING ANNUAL CPI INFLATION SERIES AND SPECIFIC CO

Perform two inflation analyses: (Series A) A country's headline Consumer 138.5, and Period 5 = 145.8. Compute the annual CPI inflation rates for P petrol rose from R19.89 to R23.22, a local taxi fare rose from R14.00 to R each item.

Verified Solution: Series A: Annual inflation rates are 5.24%, 5.30%, 5.5

$$\text{Series A Annual Rates: } \pi_2 = \frac{124.6 - 118.4}{118.4} = 5.24\%; \quad \pi$$
$$\text{Series A Four-Ye}$$
$$\text{Series B Item Increases: Petrol} = \frac{23.22 - 19.}{19.89}$$

First-Principles Proof: Correct! Each percentage change divides the per commodity price increases.

IDENTIFYING THE BASE PERIOD FROM CURRENT VS CONSTANT AGGREGATES AND COMPUTING DEFLATOR INFLATION • MEDIUM

Examine two National Accounts price-index problems: (Problem 1) In Period 1, Gross National Income (GNI) at current prices is R2,946,328 million and GNI at constant prices is also R2,946,328 million. In Period 2, GNI at current prices is R3,165,941 million and GNI at constant prices is R3,094,761 million. Explain why current and constant GNI are identical in Period 1 and compute the implicit price deflator for both periods. (Problem 2) An economy's Nominal GDP rises from R58,421 million in Period 1 to R60,798 million in Period 2, while Real GDP rises from R46,324 million to R47,760 million. Calculate the GDP deflator for both periods and the Period 2 inflation rate.

Verified Solution: Problem 1: Period 1 is the base year (Deflator = 100.00) and Period 2 Deflator = 102.30; Problem 2: Deflator rises from 126.11 to 127.30, yielding an inflation rate of 0.94%.

$$\text{Problem 1: Period 1 is the base period, so current prices equal constant base prices} \implies \text{Deflator}_1 = \frac{2,946,328}{2,946,328} \times 100 = 100.00$$

$$\text{Problem 1 Period 2 Deflator} = \frac{3,165,941}{3,094,761} \times 100 = 102.30 \quad (\pi = 2.30\%)$$

$$\text{Problem 2 Deflators: } \text{Defl}_1 = \frac{58,421}{46,324} \times 100 = 126.11; \quad \text{Defl}_2 = \frac{60,798}{47,760} \times 100 = 127.30$$

$$\text{Problem 2 Inflation: } \pi_2 = \frac{127.30 - 126.11}{126.11} \times 100 = 0.94\%$$

First-Principles Proof: Correct! Current and constant aggregates coincide in the base period (Deflator = 100.0), and in Problem 2 the GDP deflator increases from 126.11 to 127.30 (+0.94%).

CHAPTER 21 Mohr 6e, pp. 467–484 (PDF pp. 1035–1063) • Unemployment & Poverty

Chapter 21: Unemployment, Poverty and Inequality

The unemployment pool inflows & outflows, strict (narrow) vs expanded (broad) definitions, labour force participation & absorption rates, four types of unemployment (frictional, seasonal, cyclical, structural), demand vs supply-side policies, poverty lines, and SA's Triple Challenge.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Strict vs Expanded Labour Force & Unemployment Rate

$$\text{LF}_{\text{strict}} = E + U_{\text{active}}, \quad u_{\text{strict}} = \frac{U_{\text{active}}}{E + U_{\text{active}}} \times 100\%$$

Meaning: The strict definition requires active job search in the past 4 weeks; the expanded definition adds discouraged work-seekers to both the numerator and denominator.

Expanded (Broad) Unemployment Rate

$$u_{\text{broad}} = \left(\frac{U_{\text{active}} + U_{\text{discouraged}}}{E + U_{\text{active}} + U_{\text{discouraged}}} \right) \times 100\%$$

Meaning: Includes discouraged work-seekers who want work and are available to start, even if they stopped actively searching.

Labour Force Participation & Absorption Rates

$$\text{LFPR} = \frac{\text{Labour Force}}{\text{Working-Age Pop (15–64)}} \times 100\%, \quad \text{AR} = \frac{\text{Employed}}{\text{Working-Age Pop}} \times 100\%$$

Meaning: Note the denominator trap: Unemployment rate divides by the Labour Force, whereas LFPR and Absorption Rate divide by the total Working-Age Population (aged 15–64).

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

The Unemployment Pool: Four Inflows & Two Outflows (Mohr §21.1, pp. 468–470)

The level of unemployment is a stock that rises when inflows exceed outflows. People ENTER the unemployment pool for 4 reasons: (1) New entrants looking for a first job (school leavers/graduates), (2) Re-entrants returning to the labour force, (3) Job losers laid off or retrenched, and (4) Voluntary quitters searching for a better job. People LEAVE the pool by: (1) Finding employment, or (2) Withdrawing from the labour force (retiring or becoming discouraged under the strict definition).

Worked Proof: Even if an economy creates 100,000 new jobs in a quarter, the unemployment rate will increase if 250,000 new school-leavers enter the labour force simultaneously.

Four Types of Unemployment & Policy Prescriptions (Mohr §21.1, pp. 472–477)

Unemployment is classified into four types: (1) Frictional (short-term search between jobs or after graduation), (2) Seasonal (predictable off-season layoffs in agriculture/tourism, e.g. Elgin fruit pickers), (3) Cyclical / Demand-deficient (caused by recessions/downswings in aggregate demand), and (4) Structural (long-term mismatch between worker skills/location and job requirements, automation replacing travel agents, or declining industries like Rustenburg platinum mines).

Worked Proof: Cyclical unemployment is treated with demand-side expansionary monetary/fiscal policy (↓repo rate, ↑G, ↓T), whereas structural unemployment requires supply-side policies (vocational retraining, education reform, SMME deregulation, labour-intensive industrial policy).

Strict vs. Expanded Unemployment & Calculation Trap (Mohr §21.1, pp. 469–471)

Stats SA's strict (official/narrow) definition excludes discouraged work-seekers because they did not take active steps to look for work in the four weeks prior to the survey. The expanded (broad) definition includes discouraged work-seekers.

Worked Proof: In Elaria: Employed = 6,750,000, Active Unemployed = 2,220,000, Discouraged = 1,030,000 (plus students, homemakers, etc.). Broad Unemployed = 2,220,000 + 1,030,000 = 3,250,000; Broad Labour Force = 6,750,000 + 3,250,000 = 10,000,000; Broad Unemployment Rate = (3.25m / 10.0m) × 100 = 32.50%.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (The Unemployment Pool: Four Inflows & Two Outflows): Assuming a rising unemployment rate proves that zero new jobs are being created or that GDP is shrinking.

First-Principles Law: The unemployment rate rises whenever the labour force grows faster than employment creation (jobless growth)—it measures the proportion of the economically active population unable to find work.

Fallacy (Four Types of Unemployment & Policy Prescriptions): Classifying a worker permanently replaced by online algorithms or automation as 'cyclically unemployed'.

First-Principles Law: Technological displacement and skills mismatches persist even during economic booms; they represent structural (or technological) unemployment, which is South Africa's dominant challenge.

Fallacy (Strict vs. Expanded Unemployment & Calculation Trap): Including students, homemakers, pensioners, children (<15), or the medically disabled in the labour force denominator.

First-Principles Law: Students, homemakers, retirees, and children under 15 are Not Economically Active (outside the labour force) under both strict and expanded definitions.

STANDALONE ANALYTICAL PROBLEMS & DERIVATIONS (6 PROBLEMS)

CHAPTER 21 — UNEMPLOYMENT, POVERTY AND INEQUALITY — MOHR 6E, §21.1, PP. 469–471 • MEDIUM

In the economy of Elaria (Q2 Period), workforce statistics are: Employed = 6 750 000; Unemployed (actively seeking work) = 2 220 000; Discouraged work seekers = 1 030 000; Students = 2 950 000; Homemakers = 1 280 000; Too old/young to work = 740 000; Illness = 640 000; Disability = 560 000. Calculate the strict (narrow) and broad (expanded) unemployment rates.

Verified Solution: The broad unemployment rate is 32.50% (3.25m / 10.00m) and the strict rate is 24.75%.

$$u_{\text{strict}} = \frac{2\,220\,000}{6\,750\,000 + 2\,220\,000} \times 100 = \frac{2.22\text{m}}{8.97\text{m}} \times 100 = 24.75\%.$$

$$u_{\text{broad}} = \frac{2\,220\,000 + 1\,030\,000}{6\,750\,000 + 2\,220\,000 + 1\,030\,000} \times 100 = \frac{3.25\text{m}}{10.00\text{m}} \times 100 = 32.50\%.$$

First-Principles Proof: Correct! Broad unemployment adds discouraged work-seekers (1.03m) to both active unemployed (2.22m -> 3.25m) and the strict labour force (8.97m -> 10.00m), giving 32.50%.

CHAPTER 21 — UNEMPLOYMENT, POVERTY AND INEQUALITY — MOHR 6E, §21.1, PP. 469–471 • MEDIU

Labour statistics for Tigeria are as follows: Total adult population = 34 000 000; Total employed = 18 600 000; Unen seekers) = 2 500 000. Calculate Tigeria's strict labour force, broad labour force, and broad unemployment rate.

Verified Solution: Strict labour force is 27.5m, broad labour force is 30.0m, and broad unemployment is 38.00%.

$$\text{LF}_{\text{strict}} = 18.6\text{m} + 8.9\text{m} = 27.5\text{m}; \quad \text{LF}_{\text{broad}} = 27.5\text{m}$$

$$u_{\text{broad}} = \frac{8.9\text{m} + 2.5\text{m}}{30.0\text{m}} \times 100 = \frac{11.4\text{m}}{30.0\text{m}} \times 100$$

First-Principles Proof: Correct! Strict LF = 18.6m + 8.9m = 27.5m; Broad LF adds 2.5m discouraged = 30.0m; Broac

CHAPTER 21 — UNEMPLOYMENT, POVERTY AND INEQUALITY — MOHR 6E, §21.1, PP. 472–477 • EASY

Stats SA surveyed three unemployed individuals: (1) Yetunde worked on a soft-fruit farm in Elgin during the picking season and is now off-season; (2) Cipriano was a club bouncer retrenched during an economic recession and loadshedding downturn until business recovers; (3) Anouska worked at a travel agency that closed because online algorithms replaced human booking staff. Classify their unemployment types.

Verified Solution: Yetunde is seasonally unemployed, Cipriano is cyclically unemployed, and Anouska is structurally unemployed.

$$\text{Yetunde} \rightarrow \text{Seasonal}; \quad \text{Cipriano} \rightarrow \text{Cyclical}; \quad \text{Anouska} \rightarrow \text{Structural}.$$

First-Principles Proof: Correct! Agricultural harvest work is seasonal, recession/downturn layoffs are cyclical, and permanent displacement by online algorithms is structural.

CHAPTER 21 — UNEMPLOYMENT, POVERTY AND INEQUALITY — MOHR 6E, §20.6 & §21.1, PP. 459–471

In macroeconomic labour analysis, what does a rightward shift of the short-run Phillips curve indicate, and what ca increases?

Verified Solution: Every inflation level has higher unemployment; the unemployed share of the labour force rose.

$$\text{Rightward Phillips Shift} \implies \text{Stagflation (higher } \pi \text{ at ever}$$

First-Principles Proof: Correct! A rightward Phillips curve shift means a worse trade-off (more unemployment at e unemployed share of the labour force grew.

CHAPTER 21 — UNEMPLOYMENT, POVERTY AND INEQUALITY — MOHR 6E, §21.1, PP. 468–471 • EASY

Labour market survey statistics for an economy show the following: in Period 1, 14 500 000 people were employed and 1 500 000 were strictly unemployed; in Period 2, 15 000 000 people were employed and 2 500 000 were strictly unemployed. Calculate the total labour force in Period 1 and Period 2, and the official unemployment rate in Period 2.

Verified Solution: The labour force was 16.0 million in Period 1 and 17.5 million in Period 2, and Period 2 unemployment was 14.29%.

$$LF_1 = 14.5\text{m} + 1.5\text{m} = 16.0\text{m}; \quad LF_2 = 15.0\text{m} + 2.5\text{m} = 17.5\text{m}. \quad u_2 = \frac{2.5}{17.5} \times 100 = 14.29\%.$$

First-Principles Proof: Correct! Summing employed and unemployed yields LF = 16.0m (14.5m + 1.5m) in Period 1 and 17.5m (15.0m + 2.5m) in Period 2, so u = 2.5 / 17.5 × 100 = 14.29%.

FILTERING NON-WORKING-AGE COHORTS, DERIVING EMPLOYMENT COUNTS, AND CLASSIFYING UNEM

Solve three labour market problems: (Case 1) A country has a total population of 80 million, which includes 15 milli the labour force, 20 million employed workers, and 10 million unemployed job-seekers (plus 15 million other econo Unemployment Rate. (Case 2) A second economy has a working-age population of 55 million, 26 million employed Calculate the Expanded (Broad) Labour Force Participation Rate (LFPR) and Expanded Unemployment Rate. (Case 3 versus a financial advisor retrenched during a national recession.

Verified Solution: Case 1: LF = 30m, u = 33.33%; Case 2: Expanded LF = 38m (LFPR = 69.09%), Expanded u = 31.5

$$\text{Case 1: Strict Labour Force } (LF) = E + U = 20\text{m} + 10\text{m} = 30 \text{ mi}$$

$$\text{Case 2: Expanded } LF = 26\text{m} + 7\text{m} + 5\text{m} = 38 \text{ million}; \quad \text{Expande}$$

$$\text{Case 2 Expanded Unemployment Rate} = \frac{7+5}{38} \times 100 = \frac{12}{38} \times 100 = 31.58\%$$

$$\text{Case 3 Classification: Mechanisation/skill mismatch} = \text{Structural unemployment}; \quad \text{Recess}$$

First-Principles Proof: Correct! Children and inactive adults are excluded from the labour force in Case 1 (10/30 = (38/55 = 69.09%), and mechanisation vs recession represent Structural vs Cyclical unemployment.

CHAPTER 22 Mohr 6e, pp. 485–512 (PDF pp. 1064–1118) • Growth & Cycles

Chapter 22: Economic Growth and Business Cycles

Definition and measurement of real economic growth, problems with GDP as a welfare measure, the four phases of the business cycle, Classical vs Keynesian vs Structuralist cycle theories, SARB leading/coincident/lagging indicators, supply vs demand growth drivers, and capital widening vs deepening.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Annual Economic Growth Rate

$$g_t = \left(\frac{\text{Real GDP}_t - \text{Real GDP}_{t-1}}{\text{Real GDP}_{t-1}} \right) \times 100\%$$

Meaning: Economic growth MUST be calculated using Real GDP (constant prices) to eliminate price inflation; per capita real GDP adjusts further for population growth.

Rule of 70 (Doubling Time)

$$\text{Years to Double Real GDP} \approx \frac{70}{g\,(\%)}$$

Meaning: Approximates how many years it takes for real output or living standards to double at a steady annual growth rate g%.

Aggregate Production Function & Capital Intensity

$$Y = A \cdot f(K, L), \quad \text{Capital Deepening: } \uparrow\left(\frac{K}{L}\right), \quad \text{Capital Widening: } \Delta\left(\frac{K}{L}\right) = 0$$

Meaning: Capital widening equips new workers at a constant capital-to-labour ratio (K/L); capital deepening raises capital per worker, boosting labour productivity.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

Five Problems with GDP as a Measure of Welfare (Mohr §22.1, pp. 486–488)

Real GDP is flawed as a measure of societal well-being because: (1) Non-market production (e.g. unpaid household work, subsistence farming) is difficult or impossible to value, (2) Unrecorded/informal sector activity and illegal trade are omitted, (3) GDP data are frequently revised after initial estimates, (4) Aggregate or even per capita GDP ignores the distribution of income (inequality), and (5) GDP makes no distinction between economic 'goods' (hospitals, schools) and 'bads' (pollution, crime expenditure, weapons)—negative externalities are not subtracted.

Worked Proof: If Country A has twice the per capita GDP of Country B, residents of A are NOT necessarily twice as well off because per capita GDP is a simple mean that hides extreme inequality and environmental costs.

The Business Cycle: Phases, Theories & SARB Indicators (Mohr §22.2, pp. 488–494)

A complete business cycle has four phases: Trough (lowest turning point) → Expansion / Upswing (rising real GDP) → Peak / Boom (upper turning point where almost all resources are in use) → Contraction / Downswing / Recession (declining real GDP). Classical economists view the economy as inherently stable (cycles caused by exogenous shocks like weather or government meddling); Keynesian economists view cycles as endogenous (inherent to modern market economies, requiring counter-cyclical policy). SARB tracks Leading indicators (turn before the economy, e.g. share prices, building plans passed, real M1), Coincident indicators (move with the economy, e.g. manufacturing volume), and Lagging indicators (turn after, e.g. prime rate, unemployment duration).

Worked Proof: On a business cycle diagram where Point A is a Trough, Point B is a Peak, and Point C is the next Trough: A → B is the Expansion/Upswing phase, and B → C is the Contraction/Downswing phase.

Supply vs. Demand Sources of Economic Growth (Mohr §22.3, pp. 494–500)

Sustained economic growth requires both expanding productive capacity (Supply factors: natural resources/minerals, labour quantity & human capital skills, capital widening & deepening, and entrepreneurship/technology shifting the PPC and LRAS outward) and expanding aggregate expenditure (Demand factors: domestic demand C+I+G, export promotion X, and import substitution replacing Z).

Worked Proof: South Africa's mineral wealth shows that natural resources are neither necessary nor sufficient for sustained growth: commodity price volatility and exchange rate swings can create a 'resource curse' unless revenues are invested in human and physical capital.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Five Problems with GDP as a Measure of Welfare): Thinking 'GDP pertains only to production inside the country' is a welfare flaw rather than its literal geographic definition.

First-Principles Law: GDP is by definition a geographic production total; its true welfare limitations are unrecorded informal activity, non-market valuation, data revisions, inequality blindness, and counting 'bads' alongside 'goods'.

Fallacy (The Business Cycle: Phases, Theories & SARB Indicators): Confusing a slowdown in the positive rate of growth with a recession (contraction).

First-Principles Law: During a contraction (downswing), the level of real economic activity declines between a Peak and a Trough.

Fallacy (Supply vs. Demand Sources of Economic Growth): Believing that higher government spending that drives up local interest rates (crowding out private investment) always guarantees higher long-run growth.

First-Principles Law: When supply-side capacity is constrained (e.g. skills shortages or electricity bottlenecks) or higher G crowds out private investment I via higher interest rates, demand expansion fuels inflation rather than long-run real growth.

STANDALONE ANALYTICAL PROBLEMS & DERIVATIONS (4 PROBLEMS)

CHAPTER 22 — ECONOMIC GROWTH AND BUSINESS CYCLES — MOHR 6E, §22.1–22.3, PP. 485–500 • EASY

According to Chapter 22 of Mohr (6e), which phase of the business cycle occurs when almost all available resources in the economy are in use, and what does a strategy of 'import substitution' involve?

Verified Solution: The peak (boom) phase; using import tariffs to encourage local production of formerly imported goods.

$$\text{Maximum Capacity Utilisation} \rightarrow \text{Peak (Boom) Phase.} \quad \text{Import Substitution} \rightarrow \text{Tariffs/Quotas to produce former imports locally.}$$

First-Principles Proof: Correct! Resource utilisation reaches its maximum at the peak (boom) of the cycle, and import substitution uses trade protection to manufacture previously imported goods domestically.

CHAPTER 22 — ECONOMIC GROWTH AND BUSINESS CYCLES —

If per capita real GDP in Country A is twice as high as per capita real as well off, and which item is NOT a welfare limitation of GDP?

Verified Solution: Per capita GDP is an average that hides inequali

$$\text{Per Capita GDP} = \frac{\text{GDP}}{\text{Population}} \text{ (ignores Gini}$$

First-Principles Proof: Correct! Per capita GDP contains zero inform country' is simply the geographic definition of GDP (Ch 22 MCQ 5).

CHAPTER 22 — ECONOMIC GROWTH AND BUSINESS CYCLES — MOHR 6E, §22.1 & §22.3, PP. 486–498 • MEDIUM

Distinguish between 'capital widening' and 'capital deepening' in the context of long-run economic growth, and apply the Rule of 70 to determine how long it takes for a nation's real GDP per capita to double if it grows steadily at 3.5% per year.

Verified Solution: Capital widening keeps the K/L ratio constant while capital deepening raises K/L; doubling takes 20 years.

$$\text{Capital Widening: } \frac{\Delta K}{K} = \frac{\Delta L}{L} \implies \frac{K}{L} \text{ constant.} \quad \text{Capital Deepening: } \frac{K}{L} \uparrow. \quad T_{\text{double}} \approx \frac{70}{3.5} = 20 \text{ years.}$$

First-Principles Proof: Correct! Capital widening expands capital in proportion to labour growth (constant K/L), capital deepening increases capital per worker (higher K/L), and 70 / 3.5 = 20 years.

CHAPTER 22 — ECONOMIC GROWTH AND BUSINESS CYCLES —

When economists at the South African Reserve Bank (SARB) analyse and building plans passed, (2) industrial production and retail sales, construction?

Verified Solution: (1) Leading indicators (peak before the cycle), (2

$$\text{Building plans / JSE / New car sales} \rightarrow \text{Leading;} \quad \text{In}$$

First-Principles Proof: Correct! Building plans and new vehicle sales cycle, and formal employment lags behind turning points.

Comprehensive Macroeconomic Synthesis Suites & Analytical Derivations

SYNTHESIS SUITE I — NATIONAL ACCOUNTS, VALUE-ADDED & OPEN-ECONOMY MULTIPLIERS — CHAPTER 13, CHAPTER 14, CHAPTER 15, CHAPTER 16, CHAPTER 18, CHAPTER 20, CHAPTER 21

Scope: Macroeconomic Curriculum Synthesis | Verification: Quantitative & Theoretical Verification | 180 Mins • 100 Points

Conceptual Foundations — Core Definitions: Define the following macroeconomic concepts: Fiscal policy, Autonomous consumption, Comparative advantage, and Unit of account.

Analytical Solution: Fiscal policy: Government's policy in respect of the level and composition of government spending, taxation, and borrowing to influence economic activity. Autonomous consumption ($\bar{C}$): That part of consumption expenditure that is independent of the level of income (consumption when Y = 0). Comparative advantage: The ability of a country to produce a good at a lower opportunity cost (sacrificing fewer units of the alternative good) than another country. Unit of account: An agreed standard measure for stating and comparing the prices of goods and services.

Theoretical Synthesis — Cost-Push Inflation & Price Indices: Identify cost-push inflation from a leftward Aggregate Supply (AS) shift diagram, state its primary causes, and contrast the Consumer Price Index (CPI) with the Producer Price Index (PPI).

Analytical Solution: Cost-push inflation (AS shifts left/up, raising the price level P and lowering real output Y) is triggered by: wage increases exceeding productivity growth, higher imported capital/intermediate input costs, expanded profit margins, falling productivity, or natural disasters. CPI vs PPI: CPI measures the cost of living for households vs PPI measures the cost of production; CPI covers consumer goods and services vs PPI covers goods only; CPI includes VAT vs PPI excludes VAT; CPI excludes capital and intermediate goods vs PPI includes them.

Theoretical Synthesis — Labour Market Flows & Business Cycle Theories: Classify frictional, structural, cyclical, and seasonal unemployment scenarios, identify inflows and outflows of the unemployment pool, and contrast Classical versus Keynesian business cycle theories across expansion and contraction phases.

Analytical Solution: Inflows to unemployment: new entrants, re-entrants, retrenched job losers, and voluntary quitters. Outflows: finding employment or withdrawing from the labour force. Classical theory views the economy as inherently stable (cycles driven by exogenous shocks), whereas Keynesian theory views fluctuations as endogenous to unstable private investment. Upswing (Trough to Peak) features rising real GDP and employment; Downswing (Peak to Trough) features contracting real GDP.

SYNTHESIS SUITE II — REAL GROWTH, FISCAL TRANSFERS & MONETARY EQUILIBRIUM — CHAPTER 13, CHAPTER 14, CHAPTER 15, CHAPTER 16, CHAPTER 18, CHAPTER 20, CHAPTER 21, CHAPTER 22

Scope: Macroeconomic Curriculum Synthesis | Verification: Quantitative & Theoretical Verification | 180 Mins • 100 Points

Conceptual Foundations — Trade, Exchange Rates & Consumption: Distinguish between Absolute and Comparative advantage, explain the determinants of currency appreciation, describe the three characteristics of the Keynesian consumption function, and explain why autonomous government spending does not alter the size of the multiplier.

Analytical Solution: Absolute advantage compares total output from given inputs; comparative advantage compares opportunity costs. Currency appreciation is driven by stronger export demand, higher domestic interest rates attracting capital inflows, lower relative inflation, or rising commodity prices. Consumption function: $C > 0$ at $Y = 0$ ($\bar{C} > 0$), C rises with Y ($c > 0$), and $\Delta C < \Delta Y$ ($0 < c < 1$). Government spending G is autonomous (shifting A upward) whereas $\alpha = 1/[1 - c(1-t) + m]$ depends strictly on c, t, and m.

Theoretical Synthesis — Endogenous Money, Public Sector & Inequality: Explain the effect of a decline in money demand under a repo-rate monetary regime with no independent money supply curve, list the four components and three functions of the public sector, and identify the three standard measures of income inequality.

Analytical Solution: A leftward shift of money demand L along the horizontal interest rate line i_0 reduces the equilibrium quantity of money M endogenously. The public sector comprises central government, provincial government, local municipalities, and public corporations, performing allocative, distributive, and stabilisation functions. Income inequality is measured via the Lorenz curve, the Gini coefficient, and quantile ratios.

SYNTHESIS SUITE III — DEFLATORS, STATUTORY TAX TABLES & AGGREGATE EXPENDITURE — CHAPTER 13, CHAPTER 15, CHAPTER 18, CHAPTER 19, CHAPTER 20, CHAPTER 21

Scope: Macroeconomic Curriculum Synthesis | Verification: Quantitative & Theoretical Verification | 180 Mins • 100 Points

Conceptual Foundations — Stabilisation Policy & National Accounts Identity: Contrast Monetarist and Keynesian stabilisation views, explain why the income and production methods of GDP yield identical totals after a wage change, and identify the supply-side drivers of long-run economic growth.

Analytical Solution: Keynesians advocate active counter-cyclical fiscal and monetary intervention during downswings, whereas Monetarists favour stable rules. A minimum-wage change alters the distribution between wages and operating surplus without creating a discrepancy between the income and production approaches to GDP. Economic growth is driven by human capital expansion, physical capital accumulation/technology, and natural resource/institutional efficiency.

Theoretical Synthesis — Market Failure, Open-Market Operations & Disposable Income: Illustrate negative externalities (social vs private marginal cost), explain central bank open-market bond purchases, and distinguish total national income from household disposable income.

Analytical Solution: With negative externalities, Marginal Social Cost (MSC) lies above Marginal Private Cost (MPC), causing unregulated competitive markets to overproduce at a socially inefficiently low price; corrective Pigouvian taxes internalise the external cost. Open-market bond purchases inject reserves into commercial banks, expanding credit and deposits. Disposable income $Y_d = Y - T$ is income remaining after direct taxes.

SYNTHESIS SUITE IV — VALUE-ADDED CHAINS, TRADE PROTECTION & LEAKAGES — CHAPTER 13, CHAPTER 14, CHAPTER 15, CHAPTER 16, CHAPTER 17, CHAPTER 18, CHAPTER 20, CHAPTER 21, CHAPTER 22

Scope: Macroeconomic Curriculum Synthesis | Verification: Quantitative & Theoretical Verification | 180 Mins • 100 Points

Conceptual Foundations — Inflation Targeting, Phillips Curve & Trade Policy: Explain why per capita real GDP is preferred for welfare comparisons, state the central bank inflation target range (3%–6%), illustrate the short-run Phillips curve trade-off, and list the policy instruments used to restrict imports.

Analytical Solution: Real GDP per capita adjusts for both price inflation and population size. The inflation target band is 3% to 6%. The short-run Phillips curve depicts an inverse relationship between inflation and unemployment. Import restriction instruments include tariffs, quotas, domestic production subsidies, non-tariff barriers, exchange controls, and exchange-rate policy.

SYNTHESIS SUITE V — INDUCED IMPORTS, DEMOGRAPHIC FILTERING & LORENZ GEOMETRY — CHAPTER 13, CHAPTER 14, CHAPTER 15, CHAPTER 16, CHAPTER 17, CHAPTER 18, CHAPTER 20, CHAPTER 21, CHAPTER 22

Scope: Macroeconomic Curriculum Synthesis | Verification: Quantitative & Theoretical Verification | 180 Mins • 100 Points

Conceptual Foundations — Lorenz Geometry, Say's Law & Public Sector Structure: Interpret the Lorenz curve 45-degree diagonal and Gini coefficient, contrast Say's Law (supply creates its own demand) with Keynesian effective demand (demand determines output), and explain why financial cash balances are not a physical factor of production.

Analytical Solution: The 45-degree diagonal represents perfect income equality, the bowed area represents inequality, and a Gini coefficient near 0.67 indicates severe inequality. Along the 45-degree line in the Keynesian diagram, aggregate expenditure equals total income ($A = Y$); above it, $A > Y$ (inventories fall, output expands). Money is a financial medium of exchange, whereas economic capital consists strictly of physical machinery, equipment, and buildings.